

IDENTIFICATION

PRODUCT NAME: AD8E, AM8E A-D CONVERTER AND  
MULTIPLEXER DIAGNOSTIC

PRODUCT CODE: MAINDEC-8E-D68B-D-(D)

DATE CREATED: JULY 14, 1971

MAINTAINER: DIAGNOSTIC GROUP

AUTHOR: MATT TAFFEL

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READ THIS DOCUMENT PRIOR TO RUNNING PROGRAM:

## 1. ABSTRACT

This program performs basic tests on the Input/Output control logic and multiplexer. The analog tests are designed to provide a means of calibrating the converter and checking conversion parameters.

## 2. REQUIREMENTS

### 2.1 Equipment

PDP-8/E with 4K core, ASR33 teletype, AD8E A/D Converter, (AM8E Multiplexer optional), Adjustable High Quality Voltage Source (0.01% or better, Z out <1.0ohm).

NOTE: To run MONOTONICITY TEST, a wave form generator, sine or ramp, must be used.

### 2.2 Storage

Maindec resides in locations 0000-4177.

### 2.3 Preliminary Programs

All basic CPU and teletype Maindecs must have been run successfully.

NOTE: If external enable utilizing the DK8-E REAL TIME CLOCK is to be run, the Maindec for the DK8-E must be successfully run first. In addition, VC8-E Control Tests must be run prior to special LAB-E SYSTEM CHECK routine.

## 3. LOADING PROCEDURE

The binary loader is used to load the program.

#### 4. USAGE PROCEDURE

\*\*\*SEE SPECIFICATIONS FOR MAXIMUM VOLTAGE INPUTS!\*\*\*

\*\*\*INSURE THAT TELETYPE IS ON-LINE,\*\*\*

##### A. CONTROL SWITCHES

- SW0 - Suppress error messages and "END LOGIC TEST" message
- SW1 - Halt on error with PC displayed in AC.
- SW2 - Scope loop override to exit from loop on error and permit continuance of test. Also halts with converted word in AC for EXTERNAL ENABLE when there is no error.
- SW3 - Enables halt during calibration routine. Converted word is displayed in AC.
- SW4 - Must be set to run EXTERNAL ENABLE test.
- SW5 - Allows operator to explicitly select any one of the logic routines.

##### B. Normal start for control logic tests.

1. LOAD 200.
2. Press CLEAR then CONTINUE. HALT will occur.
3. Select options from switches 0,1,2,5.
4. If SW5 is present (1), select test from SW8-11.
5. Press CONTINUE.

NOTE: With SW5 down and SW2 up, any error will be reported once, then program will continue to next test.

##### C. IOT Scope Loop

1. LOAD 201.
2. Place low order six bits of IOT 65xx in SW6-11.
3. Press CLEAR, then CONTINUE.

NOTE: IOT may be reselected while running.

## D: Display Converted Value Is AC.

- 1: Apply voltage to A-D converter Input or preamplifiers.
- 2: LOAD 202.
- 3: If a HALT after conversion is desired, select SW3.
- 4: Select MPX channel from SW8-11.
- 5: Press CLEAR, then CONTINUE.
- 6: When SW3 halt select is engaged, operator may change channels, if desired, then press CONTINUE to loop. SW3 may be deselected at this time.

## E: External Enable With Real Time Clock

- 1: Apply voltage to A-D Converter Input or preamplifiers, if desired.
- 2: LOAD 203.
- 3: Set SW4.
- 4: Select switches 0 or 2 as desired.
- 5: Select channel with SW 8-11.
- 6: Press CLEAR, then CONTINUE.

NOTE: Channel may be changed while running test.

## F: Monotonicity Test

NOTE: Ramp Speed of function generator must be slower than slow rate of converter. See ENGINEERING SPECIFICATIONS.

- 1: LOAD 204
- 2: Select SW0 if desired.
- 3: Press CLEAR, then CONTINUE.
- 4: Program will halt.
- 5: Select Stall time between tests iterations by setting SW0-11. The larger the number in the switch register, the greater the stall time.

6. Press CONTINUE.

7. If error occurs, program will halt with word "N" in AC. Pressing CONTINUE will display "N+1" word in AC. Pressing CONTINUE again will restart test.

G. Resolution Accuracy Test

1. Apply a known voltage to A-D converter input.

2. LOAD 205.

3. Select SW0,1 if desired.

4. Select channel with SW8-11.

5. Press CLEAR, then CONTINUE.

6. If error occurs, program will typeout non-comparing words on TTY then continue with test.

7. If no error occurs, TTY bell will ring once then, program will recycle.

H. Successive Reads Test

1. Apply any voltage to A-D converter inputs or preamplifier.

2. LOAD 206.

3. Select SW0 if desired.

4. Select channel from SW8-11.

5. Press CLEAR, then CONTINUE.

6. If error occurs, program will halt with first read in AC. Press CONTINUE to get second read into AC.

7. To restart, press continue.

8. If no error occurs, TTY bell will ring once, then program will recycle.

J. Multiplexer noise test

1. LOAD 207.

2. Select channel in SW8-11 and apply voltage to that channel.

- 3, Select SW0 If desired.
- 4, Press CLEAR, then CONTINUE.
- 5, If error occurs, message will be typed on TTY, then routine will recycle.

#### K, LAB00E SYSTEM TEST

- 1, LOAD 205
- 2, Press CLEAR, then CONTINUE.
- 3, Program will HALT.
- 4, Load CLOCK FREQUENCY into SW3=5.
- 5, Press CONTINUE, then follow TTY Instructions.
- 6, Press CONTINUE.

#### 5, PROGRAM DESCRIPTION

##### 5.1 Control Logic Tests

- TST0 = Checks that A=D DONE and TIMING ERROR flags are cleared by Initialize.
- TST1 = Checks that A=D DONE flag can be set then cleared.
- TST2 = Checks that TIMING ERROR flag can be set then cleared.
- TST3 = Tests for unexpected Interrupt request.
- TST4 = Tests to see if ADRB Jam transfers to AC.
- TST5 = Tests to see if ADRS Jam transfers to AC.
- TST6 = Tests to see if enable register can be loaded and read back.
- TST7 = Tests to see if A=D DONE will generate Interrupt.
- TST10 = Tests to see if TIMING ERROR will generate Interrupt.

- TST11= Tests that MPX Register can be loaded and read back,
- TST12= Tests that all channels can be loaded into MPX register and read back,
- TST13= Tests auto-Increment mode of MPX register,
- TST14= Tests to see if conversion can be made in specified time,

## 5.2 Miscellaneous Tests

- A. IOT Scope Loop Test = enables IOT to be repeated for troubleshooting.
- B. External Enable Test = utilizes DK8/E Real Time Clock to start conversion. NOTE: This test can be used only if DK8/E is present in system.
- C. Display Converted Value In AC = used to calibrate converter. (See setup procedure of AD8E).>.
- D. LAB8-E SYSTEM CHECKS = assures reliability of system as homogeneous unit.

## 5.3 Analog Tests

- A. Successive Reads Tests = checks for noise in A-D buffer logic.
- B. Monotonicity Test = checks that all specified values can be converted.
- C. Resolution Accuracy Test = samples a known voltage 64 times and checks that resolution is within specification.
- D. Multiplexer Noise Test = checks for noise in MPX, ENABLE, and STATUS REGISTER.

## 6. ERROR REPORTS

### 6.1 Logic Errors

Message will be typed out once per error on teletypewriter stating test number and nature of failure.

#### 6.2 Other Errors

Message will be typed out on teletypewriter stating nature of failure.

/MAINDEC-8E-D6BB-L-(D) A TO D CONVERTER AND MULTIPLEXER DIAGNOSTIC  
 /AD8EA,AM8EA,AM8EB  
 /COPYRIGHT 1971, DIGITAL EQUIPMENT CORPORATION, MAYNARD, MASS. 01754  
 /AUTHOR: MATT TAFFEL  
 /DATE: 14 JUL 71

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/IOT DEFINITIONS

4520 ADCL= JMS I XADCL  
 4521 ADLM= JMS I XADLM  
 4522 ADST= JMS I XADST  
 4523 ADRB= JMS I XADRB  
 4524 ADSK= JMS I XADSK  
 4525 ADSE= JMS I XADSE  
 4526 ADLE= JMS I XADLE  
 4527 ADRS= JMS I XADRS  
 4530 CLOE= JMS I XCLOE  
 4531 CLSK= JMS I XCLSK  
 4532 CLZE= JMS I XCLZE  
 4533 CLSA= JMS I XCLSA  
 4534 CLED= JMS I XCLED  
 4535 CLAB= JMS I XCLAB  
 4536 DISD= JMS I XDISD  
 4537 DILX= JMS I XDILX  
 4540 DILY= JMS I XDILY  
 4541 DIXY= JMS I XDIXY  
 4542 DILE= JMS I XDILE

/CLEAR ALL  
 /LOAD MPX REG FROM AC8-11  
 /CLEAR FLAGS, START CONVERSION  
 /CLEAR DONE, READ A-D BUFFER INTO AC,  
 /SKIP ON A-D DONE, DO NOT CLEAR FLAG.  
 /SKIP ON TMG ERROR, DO NOT CLEAR FLAG.  
 /LOAD ENAB REG FROM AC 2-5. CLA,  
 /READ STATUS, ENAB, MPX REG INTO AC,  
 /AC TO CLOCK ENABLE  
 /SKIP ON CLOCK OVERFLOW  
 /ONES IN AC CLEAR CLOCK ENABLE REGISTER  
 /CLOCK STATUS TO AC, AC ONES CLEAR CLOCK STATUS REGISTER  
 /CLOCK ENABLE TO AC  
 /AC ONES TO CLOCK BUFFER  
 /SKIP ON DISPLAY DONE  
 /LOAD X  
 /LOAD Y  
 /INTENSIFY  
 /LOAD DISPLAY ENABLE FROM AC

6007 CAF= 6007  
 7002 BSW= 7002

/MPX, ENABLE, STATUS REGISTER

/ 0 AD DONE  
 / 1 TIMING ERROR  
 / 2 ENABLE INTERRUPT ON AD DONE  
 / 3 ENABLE INTERRUPT ON TIMING ERROR  
 / 4 ENABLE EXTERNAL AD START  
 / 5 AUTO-INCREMENT MODE  
 / 6,7 NOT USED  
 / 8-11 MPX CHANNEL 0-17 OCTAL

/STARTING ADDRESS

TEST

/200 NORMAL START FOR CONTROL LOGIC TESTS  
 /201 IOT SCOPE LOOP  
 /202 DISPLAY CONVERTED VALUE IN AC  
 /203 EXTERNAL ENABLE TEST  
 /204 MONOTONICITY TEST  
 /205 RESOLUTION ACCURACY TEST  
 /206 SUCCESSIVE READS TEST  
 /207 MULTIPLEXER NOISE TEST  
 /212 SYSTEM CHECK FOR LABB-E

```

0000 0020 *0
0001 0020 0
0001 5422 JMP I ,+1
0002 0020 0
0003 5424 JMP I ,+1
0004 0020 0
0005 0020 0

```

```

0017 0017 *17
0017 0145 MSGPNT, ERMSG

```

```

0020 0020 *20
0020 4020 SW0, 4020 /SWITCH REG 0 INHIBIT TYPEOUT
0021 2020 SW1, 2020 / 1 HALT ON ERROR
0022 1020 SW2, 1020 / 2 SCOPE LOOP OVERRIDE
0023 0420 SW3, 0420 / 3 CALIBRATION TEST HALT
0024 0220 SW4, 0220 / 4 EXTERNAL ENABLE
0025 0120 SW5, 0120 / 5 SELECT TEST
0026 0020 TEMP0, 0 /STORAGE BUFFER 0
0027 0020 TEMP1, 0 /STORAGE BUFFER 1
0030 0020 TEMP2, 0 / 2
0031 0020 TEMP3, 0 / 3
0032 0020 CNTR1, 0 /MONOTONICITY COUNTER
0033 0020 TALLY, 0
0034 1226 ERR, ERTYP /ERROR REPORT ROUTINE
0035 1020 XCONVT, CONV /DISPLAY CONVERTED VALUE
0036 1420 XINSTR, INSTR /IOT SCOPE LOOP
0037 2020 XMONCT, MONOT /MONOTONICITY TEST
0040 0227 K207, 207 /BELL CODE
0041 0212 K212, 212 /LINE FEED
0042 0215 K215, 215 /CARRIAGE RETURN
0043 6520 K6520, 6520
0044 7777 M1, 7777
0045 7776 M2, 7776
0046 7774 M4, 7774
0047 1020 K1020, 1020
0050 0077 K77, 0077
0051 1220 XMOVE, MOVE
0052 1024 EXTBL, EXTL
0053 2267 XSTOR, STORAG-1
0054 2420 XCOMPR, COMPAR
0055 2220 XRESOL, RESOL
0056 2051 XNOISE, NOISE
0057 2123 XGLIT, GLITCH
0060 2622 XSYST, SYST
0061 7777 ERSWIT, 7777
0062 0020 CHAN, 0
0063 1620 TAL, XTAL
0064 1647 SELECT, XSELEC
0065 1552 SETUP, XSETUP
0077 0020 CHNL, 0
0100 0021 1

```

|      |      |    |
|------|------|----|
| 0101 | 0022 | 2  |
| 0102 | 0023 | 3  |
| 0103 | 0024 | 4  |
| 0104 | 0025 | 5  |
| 0105 | 0026 | 6  |
| 0106 | 0027 | 7  |
| 0107 | 0010 | 10 |
| 0110 | 0011 | 11 |
| 0111 | 0012 | 12 |
| 0112 | 0013 | 13 |
| 0113 | 0014 | 14 |
| 0114 | 0015 | 15 |
| 0115 | 0016 | 16 |
| 0116 | 0017 | 17 |
| 0117 | 0020 | 0  |

0120 \*120

/IOT LINKS

|      |      |        |        |
|------|------|--------|--------|
| 0120 | 1410 | XADCL, | XXADCL |
| 0121 | 1414 | XADLM, | XXADLM |
| 0122 | 1420 | XADST, | XXADST |
| 0123 | 1424 | XADRB, | XXADRB |
| 0124 | 1430 | XADSK, | XXADSK |
| 0125 | 1436 | XADSE, | XXADSE |
| 0126 | 1444 | XADLE, | XXADLE |
| 0127 | 1450 | XADRS, | XXADRS |
| 0130 | 1454 | XCLOE, | XXCLOE |
| 0131 | 1460 | XCLSK, | XXCLSK |
| 0132 | 1466 | XCLZE, | XXCLZE |
| 0133 | 1472 | XCLSA, | XXCLSA |
| 0134 | 1476 | XCLED, | XXCLED |
| 0135 | 1502 | XCLAB, | XXCLAB |
| 0136 | 1506 | XDISD, | XXDISD |
| 0137 | 1514 | XDILX, | XXDILX |
| 0140 | 1520 | XDILY, | XXDILY |
| 0141 | 1524 | XDIXY, | XXDIXY |
| 0142 | 1530 | XDILE, | XXDILE |

0145 \*145

/ERROR MESSAGE LINKS

|      |      |        |        |
|------|------|--------|--------|
| 0145 | 3200 | ERMSG, | EMSG0  |
| 0146 | 3244 |        | EMSG1  |
| 0147 | 3322 |        | EMSG2  |
| 0150 | 3344 |        | EMSG3  |
| 0151 | 3372 |        | EMSG4  |
| 0152 | 3422 |        | EMSG5  |
| 0153 | 3452 |        | EMSG6  |
| 0154 | 3503 |        | EMSG7  |
| 0155 | 3540 |        | EMSG10 |
| 0156 | 3601 |        | EMSG11 |

|      |      |        |
|------|------|--------|
| 0157 | 3637 | EMSG12 |
| 0160 | 3677 | EMSG13 |
| 0161 | 3744 | EMSG14 |

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0200 0200 *200
0200 5211 JMP START /NORMAL START
0201 5436 JMP I XINSTR /IOT SCOPE LOOP OPTION
0202 5435 JMP I XCONVT /DISPLAY CONVERTED VALUE OPTION
0203 5432 JMP I EXTBL /EXTERNAL ENABLE TEST
0204 5437 JMP I XMONOT /MONOTONICITY TEST
0205 5455 JMP I XRESOL /RESOLUTION ACCURACY TEST
0206 5456 JMP I XNOISE /SUCCESSIVE READS TEST
0207 5457 JMP I XGLIT /MPX NOISE TEST
0210 5460 JMP I XSYST /LAB8-E SYSTEM CHECK
0211 7402 START, HLT
0212 7604 LAS
0213 0025 AND SW5 /SELECT SPECIFIC TEST?
0214 7440 SZA /SKIP IF NO
0215 4464 JMS I SELECT /YES

/HOUSEKEEPING
0216 7300 INITL, CLA CLL
0217 4777 JMS MESSAGE
0220 4161 XLABEL
0221 1376 TAD (144
0222 3017 DCA MSGPNT /INITIALIZE ERROR POINTER
0223 4465 JMS I SETUP
0224 6007 TST0, CAF
0225 4524 ADSK /CHECK FOR DONE FLAG - SHOULD BE CLEARED BY INIT
0226 5231 JMP ,+3
0227 4434 JMS I ERR /DONE FLAG NOT CLEARED
0230 0224 TST0
0231 4525 ADSE /CHECK FOR TIMING ERROR FLAG - SHOULD BE CLEARED BY INIT
0232 5237 JMP ,+5
0233 4434 JMS I ERR /TIMING ERROR FLAG NOT CLEARED
0234 0224 TST0
0235 5240 JMP TST1-1
0236 5224 JMP TST0
0237 4463 JMS I TAL

/CHECK TO SET DONE FLAG AND CLEAR DONE FLAG
0240 4465 JMS I SETUP
0241 7200 TST1, CLA
0242 4522 ADST /CONVERT, RESULTS NOT NEEDED
0243 1177 TAD [-100
0244 3026 DCA TEMP0
0245 2026 ISZ TEMP0
0246 5245 JMP , -1
0247 4524 ADSK
0250 7410 SKP
0251 5255 JMP ,+4
0252 4434 JMS I ERR /FLAG NOT SET
0253 0241 TST1
0254 5265 JMP TST2-1
0255 4520 ADCL /CLEAR FLAG
0256 4524 ADSK /CHECK FOR FLAG
0257 5264 JMP ,+5 /FLAG CLEARED

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0260 4434      JMS I   ERR           /FLAG NOT CLEARED
0261 0241      TST1
0262 5265      JMP      TST2-1
0263 5241      JMP      TST1
0264 4463      JMS I   TAL

/CHECK TO SET TIMING ERROR FLAG AND CLEAR TIMING ERROR FLAG
0265 4465      JMS I   SETUP
0266 7200      TST2,  CLA
0267 4522      ADST           /TWO A-D STARTS TO PRODUCE TIMING ERROR
0270 4522      ADST
0271 4525      ADSE           /CHECK FOR TIMING ERROR FLAG
0272 7410      SKP
0273 5276      JMP      .+3
0274 4434      JMS I   ERR           /FLAG NOT SET
0275 0266      TST2
0276 4520      ADCL           /CLEAR FLAG
0277 4525      ADSE           /CHECK FLAG
0300 5305      JMP      .+5
0301 4434      JMS I   ERR           /FLAG NOT CLEARED
0302 0266      TST2
0303 5306      JMP      TST3-1
0304 5266      JMP      TST2
0305 4463      JMS I   TAL

/TEST FOR UNEXPECTED INTERRUPT REQUEST
0306 4465      JMS I   SETUP
0307 7200      TST3,  CLA
0310 1176      TAD      [TST3S
0311 3004      DCA      4
0312 1317      TAD      .+5           /ERROR TRAP
0313 3001      DCA      1
0314 6001      ION           /TURN INT ON
0315 7000      NOP
0316 5322      JMP      .+4
0317 4434      JMS I   ERR           /UNEXPECTED INTERRUPT OCCURRED
0320 0307      TST3
0321 5326      JMP      TST4-1
0322 6002      TST3S,  IOF           /TURN INT OFF
0323 7410      SKP
0324 5307      JMP      TST3
0325 4463      JMS I   TAL

/TEST THAT ADRB JAM TRANSFERS TO AC
0326 4465      JMS I   SETUP
0327 7240      TST4,  CLA CMA           /AC=7777
0330 4523      ADRB           /SHOULD CLEAR AC
0331 3027      DCA      TEMPA           /SAVE AC
0332 7040      CMA
0333 4523      ADRB           /READ WITH AC=0
0334 7041      CIA
0335 1027      TAD      TEMPA
0336 7440      SZA           /EQUAL?
0337 7410      SKP

```

|      |      |       |        |           |
|------|------|-------|--------|-----------|
| 0340 | 5345 | JMP   | .*5    |           |
| 0341 | 4434 | JMS I | ERR    | /NO-ERROR |
| 0342 | 2327 | TST4  |        |           |
| 0343 | 5346 | JMP   | TST5-1 |           |
| 0344 | 5327 | JMP   | TST4   |           |
| 0345 | 4463 | JMS I | TAL    |           |

/TEST THAT ADRS JAM TRANSFERS TO AC

|      |       |         |        |                          |
|------|-------|---------|--------|--------------------------|
| 0346 | 4465  | JMS I   | SETUP  |                          |
| 0347 | 4520  | TST5,   | ADCL   |                          |
| 0350 | 4521  |         | ADLM   |                          |
| 0351 | 4522  |         | ADST   |                          |
| 0352 | 4524  |         | ADSK   |                          |
| 0353 | 5352  | JMP     | .-1    |                          |
| 0354 | 7340  | CLA CMA | CLL    | /AC=7777                 |
| 0355 | 4527  | ADRS    |        |                          |
| 0356 | 3027  | DCA     | TEMPA  | /SAVE AC, SHOULD BE 4000 |
| 0357 | 1027  | TAD     | TEMPA  |                          |
| 0360 | 7024  | RAL     |        |                          |
| 0361 | 7440  | SZA     |        | /DID ADRS CLEAR AC?      |
| 0362 | 7410  | SKP     |        |                          |
| 0363 | 5370  | JMP     | .*5    |                          |
| 0364 | 4434  | JMS I   | ERR    | /NO                      |
| 0365 | 0347  | TST5    |        |                          |
| 0366 | 5775' | JMP     | TST6-1 |                          |
| 0367 | 5347  | JMP     | TST5   |                          |
| 0370 | 4463  | JMS I   | TAL    |                          |
| 0371 | 5775' | JMP     | TST6-1 |                          |

|      |      |
|------|------|
| 0375 | 0400 |
| 0376 | 0144 |
| 0377 | 1274 |
|      | 0400 |

PAGE

/CHECKS THAT ENABLE REGISTER CAN BE LOADED AND READ BACK

|      |      |       |         |                            |
|------|------|-------|---------|----------------------------|
| 0400 | 4465 | JMS I | SETUP   |                            |
| 0401 | 7300 | TST6, | CLA CLL |                            |
| 0402 | 1175 |       | TAD     | [17                        |
| 0403 | 7022 |       | BSW     | /GET BITS AND              |
| 0404 | 4526 |       | ADLE    | /PLACE IN AC 2-5           |
| 0405 | 7440 |       | SZA     | /LOAD                      |
| 0406 | 7410 |       | SKP     |                            |
| 0407 | 5212 | JMP   | .*3     |                            |
| 0410 | 4434 | JMS I | ERR     | /AC NOT CLEARED BY ADLE    |
| 0411 | 0401 | TST6  |         |                            |
| 0412 | 7040 | CMA   |         |                            |
| 0413 | 4527 | ADRS  |         | /READ BACK                 |
| 0414 | 7022 | BSW   |         |                            |
| 0415 | 1174 | TAD   | [7761   | /CHECK FOR ONLY AC 2-5 SET |
| 0416 | 7440 | SZA   |         |                            |
| 0417 | 7410 | SKP   |         |                            |
| 0420 | 5225 | JMP   | .*5     |                            |
| 0421 | 4434 | JMS I | ERR     | /WRONG BITS                |

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0422 0401      TST6
0423 5226      JMP      TST7-1
0424 5201      JMP      TST6
0425 4463      JMS I    TAL          /DONE?

/GENERATE INTERRUPT WITH A-D DONE FLAG
0426 4465      JMS I    SETUP
0427 7200      TST7,   CLA
0430 4522      ADST          /CONVERT
0431 4524      ADSK          /DONE?
0432 5231      JMP      , -1      /WAIT
0433 1173      TAD      [DON1
0434 3002      DCA      2          /RETURN POINTER
0435 1047      TAD      K1000
0436 4526      ADLE          /LOAD INTERRUPT ENABLE
0437 6001      ION
0440 7000      NOP
0441 6002      IOF
0442 4434      JMS I    ERR          /DID NOT INTERRUPT
0443 0427      TST7
0444 5251      JMP      TST10-1
0445 4520      DON1,   ADCL          /CLEAR WORLD
0446 7410      SKP
0447 5227      JMP      TST7
0450 4463      JMS I    TAL

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/GENERATE INTERRUPT WITH TIMING ERROR FLAG
0451 4465      JMS I    SETUP
0452 7200      TST10,  CLA
0453 1172      TAD      [TMG1
0454 3002      DCA      2
0455 4522      ADST          /CAUSE ERROR HERE
0456 4522      ADST
0457 4524      ADSK          /DONE?
0460 5257      JMP      , -1
0461 4525      ADSE          /TIMING ERROR?
0462 5261      JMP      , -1
0463 7300      CLA CLL
0464 1047      TAD      K1000
0465 7010      RAR
0466 4526      ADLE          /LOAD INTERRUPT ENABLE
0467 6001      ION          /INT ON
0470 7000      NOP
0471 6002      IOF          /INT OFF
0472 4434      JMS I    ERR          /DID NOT INTERRUPT
0473 0452      TST10
0474 5301      JMP      TST11-1
0475 4520      TMG1,   ADCL          /CLEAR WORLD
0476 7410      SKP
0477 5252      JMP      TST10
0500 4463      JMS I    TAL

```

/LOAD AND READ MPX REG

```

0501 4465      JMS I  SETUP
0502 7240      TST11, CLA CMA
0503 4521      ADLM
0504 7450      SNA                      /CHECK IF AC CLEARED
0505 5311      JMP      ,+4
0506 4434      JMS I  ERR                      /AC WAS NOT CLEARED BY ADLM
0507 0502      TST11
0510 7200      CLA
0511 4521      ADLM                      /LOAD MPX REG WITH 00
0512 4527      ADRS                      /READ MPX REG
0513 0175      AND      [17]              /MASK FOR MPX REG
0514 7440      SZA
0515 7410      SKP
0516 5321      JMP      ,+3
0517 4434      JMS I  ERR                      /MPX REG NOT 0
0520 0502      TST11
0521 7040      CMA
0522 0175      AND      [17]
0523 4521      ADLM                      /MPX REG SET TO 17
0524 4527      ADRS                      /READ MPX REG
0525 0175      AND      [17]
0526 1171      TAO      [7760]            /MASK
0527 7040      CMA
0530 7440      SZA
0531 7410      SKP
0532 5337      JMP      ,+5
0533 4434      JMS I  ERR                      /MPX REG NOT 17
0534 0502      TST11
0535 5777'     JMP      TST12-1
0536 5302      JMP      TST11
0537 4463      JMS I  TAL
0540 5777'     JMP      TST12-1

```

0577 0600  
0600

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/LOAD MPX REG WITH EACH CHANNEL

```

0600 4465      JMS I  SETUP
0601 7300      TST12, CLA CLL
0602 3026      DCA      TEMP0
0603 1026      TAO      TEMP0              /CHANNEL INTO AC
0604 7040      CMA
0605 3027      DCA      TEMP1              /COMPLEMENTED CHANNEL
0606 1026      TAO      TEMP0
0607 4521      ADLM
0610 4527      ADRS                      /LOAD IT
0611 0175      AND      [17]              /READ MPX REG
0612 3030      DCA      TEMP1              /MASK 8-11
0613 1027      TAO      TEMP1              /STORE IT
0614 1030      TAO      TEMP1              /CHECK IT
0615 7001      IAC
0616 7440      SZA
0617 7410      SKP
0620 5224      JMP      ,+4
0621 4434      JMS I  ERR                      /WRONG CHANNEL

```

|      |      |         |                          |
|------|------|---------|--------------------------|
| 0622 | 0601 | TST12   |                          |
| 0623 | 5236 | JMP     | TST13-1                  |
| 0624 | 1171 | TAD     | [7760                    |
| 0625 | 1026 | TAD     | TEMP2                    |
| 0626 | 7021 | IAC     |                          |
| 0627 | 7440 | SZA     | /DONE WITH ALL CHANNELS? |
| 0630 | 7410 | SKP     | /NO                      |
| 0631 | 5235 | JMP     | ..+4                     |
| 0632 | 7300 | CLA CLL | /YES                     |
| 0633 | 2026 | ISZ     | TEMP3                    |
| 0634 | 5203 | JMP     | TST12+2                  |
| 0635 | 4463 | JMS I   | TAL                      |

/DONE WITH ALL CHANNELS?  
/NO  
/YES  
/SET NEXT CHANNEL  
/DONE WITH TEST?

/AUTO-INCREMENT MODE TEST

|      |      |         |         |
|------|------|---------|---------|
| 0636 | 4465 | JMS I   | SETUP   |
| 0637 | 7300 | TST13,  | CLA CLL |
| 0640 | 1170 | TAD     | [76     |
| 0641 | 3010 | DCA     | 10      |
| 0642 | 4520 | ADCL    |         |
| 0643 | 7200 | AUTO1,  | CLA     |
| 0644 | 1410 | TAD I   | 10      |
| 0645 | 3027 | DCA     | TEMPA   |
| 0646 | 1027 | TAD     | TEMPA   |
| 0647 | 7040 | CMA     |         |
| 0650 | 3030 | DCA     | TEMPB   |
| 0651 | 1025 | TAD     | SW5     |
| 0652 | 4526 | ADLE    |         |
| 0653 | 1027 | TAD     | TEMPA   |
| 0654 | 4521 | ADLM    |         |
| 0655 | 4522 | ADST    |         |
| 0656 | 4524 | ADSK    |         |
| 0657 | 5256 | JMP     | ..-1    |
| 0660 | 4527 | ADRS    |         |
| 0661 | 0175 | AND     | [17     |
| 0662 | 3031 | DCA     | TEMPC   |
| 0663 | 1027 | TAD     | TEMPA   |
| 0664 | 1174 | TAD     | [7761   |
| 0665 | 7640 | SZA CLA |         |
| 0666 | 5272 | JMP     | ..+4    |
| 0667 | 1410 | TAD I   | 10      |
| 0670 | 1031 | TAD     | TEMPC   |
| 0671 | 5274 | JMP     | AUTO2   |
| 0672 | 1031 | TAD     | TEMPC   |
| 0673 | 1230 | TAD     | TEMPB   |
| 0674 | 7440 | AUTO2,  | SZA     |
| 0675 | 7410 | SKP     |         |
| 0676 | 5302 | JMP     | ..+4    |
| 0677 | 4434 | JMS I   | ERR     |
| 0700 | 0637 | TST13   |         |
| 0701 | 5310 | JMP     | TST14-1 |
| 0702 | 1031 | TAD     | TEMPC   |
| 0703 | 7440 | SZA     |         |
| 0704 | 5243 | JMP     | AUTO1   |
| 0705 | 7410 | SKP     |         |

/CHANNEL N

/AUTO-INCREMENT BIT  
/LOAD ENABLE REG  
/CHANNEL N  
/LOAD MPX REG  
/START CONVERSION  
/WAIT FOR  
/DONE FLAG  
/READ STATUS  
/MASK OUT ALL BUT MPX REG

/CHECK IF CHANNEL 17 INCREMENTED TO 0

/IF CHANNEL 17 SKIP

/AC SHOULD = 0

/CHECK FOR CHANNEL N+1

/WRONG CHANNEL

/LAST CHANNEL?  
/NO

0706 5237 JMP TST13  
0707 4463 JMS I TAL

/ROUTINE TO CHECK THAT CONVERSION CAN BE MADE IN 20 MICROSECS

0710 4465 JMS I SETUP  
0711 7300 TST14, CLA CLL  
0712 1377 TAD (-6  
0713 3026 DCA TEMP0  
0714 4520 ADCL  
0715 4522 ADST  
0716 2026 ISZ TEMP0  
0717 5316 JMP .-1  
0720 4524 ADSE  
0721 7410 SKP  
0722 5327 JMP .+5  
0723 4434 JMS I ERR  
0724 0711 TST14  
0725 5330 JMP FINIS  
0726 5311 JMP TST14  
0727 4463 JMS I TAL  
0730 7604 FINIS, LAS  
0731 0020 AND SW0  
0732 7640 SZA CLA  
0733 5337 JMP .+4  
0734 4776 JMS MESSAGE  
0735 4146 XEND  
0736 7200 CLA  
0737 1040 TAD K207  
0740 4775 JMS PRLP  
0741 5774 JMP TST0-3

/TIME OUT ERROR

/SWITCH SET TO DELETE  
/TYPEOUT OF END LOGIC TEST

/RETURN TO BEGINNING OF LOGIC TESTS,

0774 0221  
0775 1534  
0776 1274  
0777 7772  
1000

PAGE

/ROUTINE TO DISPLAY CONVERTED VALUE IN AC.

CONVT, ADCL /CLEAR WORLD  
1001 3026 DCA TEMP0  
1002 7604 LAS /LOAD CHANNEL  
1003 4521 ADLM /LOAD MPX REG  
1004 4522 ADST /CONVERT  
1005 4524 ADSE /DONE?  
1006 5205 JMP .-1 /WAIT  
1007 4523 ADSE /READ A-D BUFFER  
1010 2026 ISZ TEMP2 /STALL TO DISPLAY  
1011 5210 JMP .-1 /CONVERTED VALUE  
1012 2026 ISZ TEMP0 /IN AC FOR  
1013 5212 JMP .-1 /33 MILLISECONDS  
1014 3031 DCA TEMP0  
1015 7624 LAS /CHECK IF HALT DESIRED  
1016 0023 AND SW3

|      |      |         |                                       |
|------|------|---------|---------------------------------------|
| 1017 | 7650 | SNA CLA |                                       |
| 1020 | 5223 | JMP     | ,+3                                   |
| 1021 | 1031 | TAD     | TEMPC                                 |
| 1022 | 7402 | HLT     | /PRESS CONTINUE IF NOT DONE ADJUSTING |
| 1023 | 5220 | JMP     | CONVT /LOOP                           |

/ROUTINE TO CHECK FOR EXTERNAL ENABLE FROM REAL TIME CLOCK

|      |      |        |         |       |                                     |
|------|------|--------|---------|-------|-------------------------------------|
| 1024 | 4465 | EXTL,  | JMS I   | SETUP |                                     |
| 1025 | 4520 |        | ADCL    |       | /CLEAR ALL                          |
| 1026 | 7604 |        | LAS     |       |                                     |
| 1027 | 0024 |        | AND     | SW4   | /CHECK FOR EXT'L ENABLE SWITCH      |
| 1030 | 7450 |        | SNA     |       |                                     |
| 1031 | 7402 |        | HLT     |       | /SWITCH NOT SET, STOP!              |
| 1032 | 7604 | EXT1,  | LAS     |       |                                     |
| 1033 | 0024 |        | AND     | SW4   |                                     |
| 1034 | 4526 |        | ADLE    |       |                                     |
| 1035 | 7604 |        | LAS     |       |                                     |
| 1036 | 0175 |        | AND     | [17   |                                     |
| 1037 | 4521 |        | ADLM    |       | /LOAD CHANNEL FROM SW8-11           |
| 1040 | 1377 |        | TAD     | (4340 | /LOAD CLOCK ENABLE REG              |
| 1041 | 4530 |        | CLOE    |       | /TRIGGER FROM RTC                   |
| 1042 | 7040 |        | CMA     |       |                                     |
| 1043 | 4532 |        | CLZE    |       |                                     |
| 1044 | 4531 |        | CLSK    |       | /OCCURS ON OVERFLOW                 |
| 1045 | 5244 |        | JMP     | .-1   |                                     |
| 1046 | 4533 |        | CLSA    |       | /STOP CLOCK                         |
| 1047 | 7240 |        | CLA CMA |       |                                     |
| 1050 | 4532 |        | CLZE    |       |                                     |
| 1051 | 7200 |        | CLA     |       |                                     |
| 1052 | 2026 |        | ISZ     | TEMP2 | /TIME OUT                           |
| 1053 | 5252 |        | JMP     | .-1   |                                     |
| 1054 | 4524 |        | ADSK    |       |                                     |
| 1055 | 4776 |        | JMS     | ERPT3 | /CONVERSION NOT MADE                |
| 1056 | 4523 |        | ADRB    |       |                                     |
| 1057 | 3027 |        | DCA     | TEMPA | /STORE CONVERSION                   |
| 1060 | 7604 |        | LAS     |       |                                     |
| 1061 | 0022 |        | AND     | SW2   | /LOOP?                              |
| 1062 | 7650 |        | SNA CLA |       |                                     |
| 1063 | 5266 |        | JMP     | EXTTE | /YES                                |
| 1064 | 1027 |        | TAD     | TEMPA | /HALT WITH CONVERTED                |
| 1065 | 7402 |        | HLT     |       | /VALUE IN AC.                       |
| 1066 | 4465 | EXTTE, | JMS I   | SETUP |                                     |
| 1067 | 4520 |        | ADCL    |       |                                     |
| 1070 | 7604 |        | LAS     |       |                                     |
| 1071 | 0024 |        | AND     | SW4   |                                     |
| 1072 | 4526 |        | ADLE    |       |                                     |
| 1073 | 7240 |        | CLA CMA |       |                                     |
| 1074 | 4535 |        | CLAB    |       | /CLOCK BUFFER = 7777                |
| 1075 | 7200 |        | CLA     |       |                                     |
| 1076 | 1375 |        | TAD     | (164) | /TO GIVE TIMING ERROR ON NEXT CLOCK |
| 1077 | 4530 |        | CLOE    |       |                                     |
| 1100 | 7000 |        | NOP     |       |                                     |

|      |       |      |       |                         |
|------|-------|------|-------|-------------------------|
| 1101 | 4525  | ADSE |       | /TIMING ERROR SEEN HERE |
| 1122 | 4776' | JMS  | ERPT3 | /DID NOT RAISE FLAG     |
| 1123 | 7240  | CLA  | CMA   |                         |
| 1124 | 4532  | CLPE |       |                         |
| 1125 | 7200  | CLA  |       |                         |
| 1126 | 4520  | ADCL |       |                         |
| 1107 | 1024  | TAD  | SW4   |                         |
| 1110 | 4526  | ADLE |       |                         |
| 1111 | 7240  | CLA  | CMA   |                         |
| 1112 | 4535  | CLAB |       |                         |
| 1113 | 7200  | CLA  |       |                         |
| 1114 | 4535  | CLAR |       |                         |
| 1115 | 1374  | TAD  | (-6   |                         |
| 1116 | 3031  | DCA  | TEMPC |                         |
| 1117 | 2031  | ISZ  | TEMPC |                         |
| 1120 | 5317  | JMP  | .-1   |                         |
| 1121 | 4524  | ADSK |       |                         |
| 1122 | 5325  | JMP  | .-3   |                         |
| 1123 | 4776' | JMS  | ERPT3 |                         |
| 1124 | 4520  | ADCL |       |                         |
| 1125 | 7200  | CLA  |       |                         |
| 1126 | 1040  | TAD  | K207  |                         |
| 1127 | 4773' | JMS  | PRLP  |                         |
| 1130 | 5232  | JMP  | EXT1  |                         |

1173 1534  
1174 7772  
1175 1640  
1176 1732  
1177 4340  
1200

PAGE

/SUBROUTINE TO MOVE VARIABLE LENGTH DATA FIELDS

|      |      |       |       |                         |
|------|------|-------|-------|-------------------------|
| 1200 | 0000 | MOVE, | 0     |                         |
| 1201 | 7300 | CLA   | CLL   |                         |
| 1202 | 1600 | TAD   | I     | MOVE                    |
| 1203 | 3223 | DCA   | FADDR | /GET "FROM ADDR" AND    |
| 1204 | 2200 | ISZ   | MOVE  | /STORE                  |
| 1205 | 1600 | TAD   | I     | MOVE                    |
| 1206 | 3224 | DCA   | TADDR | /GET "TO ADDR" AND      |
| 1207 | 2200 | ISZ   | MOVE  | /STORE                  |
| 1210 | 1600 | TAD   | I     | MOVE                    |
| 1211 | 3225 | DCA   | MCTR  | /GET "MOVE COUNT" AND   |
| 1212 | 2200 | ISZ   | MOVE  | /STORE                  |
| 1213 | 7200 | CLAB  |       | /SETUP FOR EXIT         |
| 1214 | 1623 | TAD   | I     | FADDR                   |
| 1215 | 3624 | DCA   | I     | TADDR                   |
| 1216 | 2223 | ISZ   | FADDR | /GET "FROM" WORD        |
| 1217 | 2224 | ISZ   | TADDR | /STORE AT "TO" LOCATION |
| 1220 | 2225 | ISZ   | MCTR  | /+1 TO "FROM" ADDR      |
| 1221 | 5213 | JMP   | MOVEA | /+1 TO "TO" ADDR        |
| 1222 | 5600 | JMP   | I     | MOVE                    |

1223 0000 FADDR, 2  
 1224 0000 TADDR, 2  
 1225 0000 MCTR, 2

## /ERROR TYPEOUT ROUTINE

1226 0000 ERTYP, 0  
 1227 7200 CLA  
 1230 1346 TAD IND  
 1231 7640 SZA CLA  
 1232 5243 JMP EOUT+1 /TYPE ERROR MESSAGE ONE TIME ONLY  
 1233 7604 LAS  
 1234 0000 AND SW0 /SUPPRESS TYPEOUT?  
 1235 7710 SPA CLA  
 1236 5247 JMP EOUT+5 /YES  
 1237 1417 TAD I MSGPNT /GET POINTER FOR ERROR MESSAGE  
 1240 3242 DCA EOUT  
 1241 4274 JMS MESSAGE  
 1242 7402 EOUT, HLT  
 1243 7200 CLA  
 1244 1346 TAD IND  
 1245 7640 SZA CLA  
 1246 5250 JMP .+2  
 1247 2346 ISZ IND  
 1250 7604 LAS  
 1251 0001 AND SW1 /HALT ON ERROR SWITCH ON?  
 1252 7650 SNA CLA /SKIP IF ON  
 1253 5257 JMP SCOPE  
 1254 1226 TAD ERTYP  
 1255 1044 TAD M1  
 1256 7402 HLT /HALT WITH ERROR P.C. IN AC.  
 1257 7604 SCOPE, LAS  
 1260 0002 AND SW2 /OVERRIDE LOOP?  
 1261 7640 SZA CLA  
 1262 5272 JMP .+10  
 1263 1626 TAD I ERTYP /NO  
 1264 3271 DCA EXIT  
 1265 1017 TAD MSGPNT  
 1266 1044 TAD M1  
 1267 3017 DCA MSGPNT  
 1270 5671 JMP I EXIT  
 1271 7422 EXIT, HLT  
 1272 2226 ISZ ERTYP /YES  
 1273 5626 JMP I ERTYP

## /MESSAGE ROUTINE FOR LOGIC ERRORS

1274 0000 MESSAGE, 0  
 1275 7240 CLA CMA  
 1276 1674 TAD I MESSAGE  
 1277 3012 DCA 12  
 1300 2274 ISZ MESSAGE

|      |      |           |           |
|------|------|-----------|-----------|
| 1301 | 1410 | TAD I     | 10        |
| 1302 | 3313 | DCA       | MSRGHT    |
| 1303 | 1313 | TAD       | MSRGHT    |
| 1304 | 7212 | DTR       |           |
| 1305 | 7012 | RTR       |           |
| 1306 | 7212 | RTR       |           |
| 1307 | 4314 | JMS       | TYPECH    |
| 1310 | 1313 | TAD       | MSRGHT    |
| 1311 | 4314 | JMS       | TYPECH    |
| 1312 | 5321 | JMP       | MESSAGE+5 |
| 1313 | 0020 | MSRGHT, 0 |           |
| 1314 | 0020 | TYPECH, 0 |           |
| 1315 | 0050 | AND       | K77       |
| 1316 | 7452 | SNA       |           |
| 1317 | 5674 | JMP I     | MESSAGE   |
| 1320 | 1377 | TAD       | (-40      |
| 1321 | 7510 | SPA       |           |
| 1322 | 5325 | JMP       | .+3       |
| 1323 | 1376 | TAD       | (240      |
| 1324 | 5340 | JMP       | MTP       |
| 1325 | 7021 | IAC       |           |
| 1326 | 7442 | SZA       |           |
| 1327 | 5332 | JMP       | .+3       |
| 1330 | 1042 | TAD       | K215      |
| 1331 | 5342 | JMP       | MTP       |
| 1332 | 7021 | IAC       |           |
| 1333 | 7442 | SZA       |           |
| 1334 | 5337 | JMP       | .+3       |
| 1335 | 1041 | TAD       | K212      |
| 1336 | 5342 | JMP       | MTP       |
| 1337 | 1375 | TAD       | (336      |
| 1340 | 6046 | MTP, TLS  |           |
| 1341 | 6041 | TSF       |           |
| 1342 | 5341 | JMP       | .-1       |
| 1343 | 6042 | TCF       |           |
| 1344 | 7220 | CLA       |           |
| 1345 | 5714 | JMP I     | TYPECH    |
| 1346 | 0020 | IND, 0    |           |

1375 0336  
1376 0240  
1377 7742  
1420

PAGE  
/SCOPE LOOP FOR IOTS 65XX.  
INSTR, NOP

|      |      |     |         |
|------|------|-----|---------|
| 1400 | 7020 | LAS |         |
| 1401 | 7604 | AND | K77     |
| 1402 | 0050 | TAD | K6520   |
| 1403 | 1043 | DCA | .+1     |
| 1404 | 3225 | HLT |         |
| 1405 | 7402 | NOP |         |
| 1406 | 7220 | NOP |         |
| 1407 | 5211 | JMP | INSTR+1 |

/SELECT IOT FROM SR 6-11  
/MASK OUT AC 2-5  
/CREATE IOT  
/LOCATION OF IOT  
/POSSIBLE SKIP  
/LOOP

## /IOT SUBROUTINES

|      |      |           |        |                         |
|------|------|-----------|--------|-------------------------|
| 1410 | 0000 | XXADCL, 0 |        |                         |
| 1411 | 6530 | 6530      |        | /CLEAR ALL              |
| 1412 | 5610 | JMP I     | XXADCL |                         |
| 1413 | 7402 | HLT       |        |                         |
| 1414 | 0000 | XXADLM, 0 |        |                         |
| 1415 | 6531 | 6531      |        | /LOAD MPX REG           |
| 1416 | 5614 | JMP I     | XXADLM |                         |
| 1417 | 7402 | HLT       |        |                         |
| 1420 | 0000 | XXADST, 0 |        |                         |
| 1421 | 6532 | 6532      |        | /START CONVERSION       |
| 1422 | 5620 | JMP I     | XXADST |                         |
| 1423 | 7402 | HLT       |        |                         |
| 1424 | 0000 | XXADRB, 0 |        |                         |
| 1425 | 6533 | 6533      |        | /READ A-D BUFFER        |
| 1426 | 5624 | JMP I     | XXADRB |                         |
| 1427 | 7402 | HLT       |        |                         |
| 1430 | 0000 | XXADSK, 0 |        |                         |
| 1431 | 6534 | 6534      |        | /SKIP ON A-D DONE       |
| 1432 | 7410 | SKP       |        |                         |
| 1433 | 2230 | ISZ       | XXADSK |                         |
| 1434 | 5630 | JMP I     | XXADSK |                         |
| 1435 | 7402 | HLT       |        |                         |
| 1436 | 0000 | XXADSE, 0 |        |                         |
| 1437 | 6535 | 6535      |        | /SKIP ON TIMING ERROR   |
| 1440 | 7410 | SKP       |        |                         |
| 1441 | 2236 | ISZ       | XXADSE |                         |
| 1442 | 5636 | JMP I     | XXADSE |                         |
| 1443 | 7402 | HLT       |        |                         |
| 1444 | 0000 | XXADLE, 0 |        |                         |
| 1445 | 6536 | 6536      |        | /LOAD ENABLE REGISTER   |
| 1446 | 5644 | JMP I     | XXADLE |                         |
| 1447 | 7402 | HLT       |        |                         |
| 1450 | 0000 | XXADRS, 0 |        |                         |
| 1451 | 6537 | 6537      |        | /READ STATUS REGISTER   |
| 1452 | 5650 | JMP I     | XXADRS |                         |
| 1453 | 7402 | HLT       |        |                         |
| 1454 | 0000 | XXCLOE, 0 |        |                         |
| 1455 | 6132 | 6132      |        | /LOAD CLOCK ENABLE      |
| 1456 | 5654 | JMP I     | XXCLOE |                         |
| 1457 | 7402 | HLT       |        |                         |
| 1460 | 0000 | XXCLSK, 0 |        |                         |
| 1461 | 6131 | 6131      |        | /SKIP ON CLOCK OVERFLOW |
| 1462 | 7410 | SKP       |        |                         |
| 1463 | 2260 | ISZ       | XXCLSK |                         |

|      |      |           |        |                                                 |
|------|------|-----------|--------|-------------------------------------------------|
| 1464 | 5660 | JMP I     | XXCLSK |                                                 |
| 1465 | 7402 | HLT       |        |                                                 |
| 1466 | 0002 | XXCLZE, 0 |        |                                                 |
| 1467 | 6130 | 6130      |        | /ONES IN AC CLEAR CLOCK ENABLE REG              |
| 1470 | 5666 | JMP I     | XXCLZE |                                                 |
| 1471 | 7402 | HLT       |        |                                                 |
| 1472 | 0000 | XXCLSA, 0 |        |                                                 |
| 1473 | 6135 | 6135      |        | /CLOCK STATUS TO AC, AC ONES CLR CLK STATUS REG |
| 1474 | 5672 | JMP I     | XXCLSA |                                                 |
| 1475 | 7402 | HLT       |        |                                                 |
| 1476 | 0000 | XXCLED, 0 |        |                                                 |
| 1477 | 6134 | 6134      |        | /CLOCK ENABLE TO AC                             |
| 1500 | 5676 | JMP I     | XXCLED |                                                 |
| 1501 | 7402 | HLT       |        |                                                 |
| 1502 | 0000 | XXCLAB, 0 |        |                                                 |
| 1503 | 6133 | 6133      |        | /AC ONES TO CLOCK BUFFER                        |
| 1504 | 5702 | JMP I     | XXCLAB |                                                 |
| 1505 | 7402 | HLT       |        |                                                 |
| 1506 | 0000 | XXDISD, 0 |        |                                                 |
| 1507 | 6052 | 6052      |        | /SKIP ON DISPLAY DONE                           |
| 1510 | 7410 | SKP       |        |                                                 |
| 1511 | 2306 | ISZ       | XXDISD |                                                 |
| 1512 | 5706 | JMP I     | XXDISD |                                                 |
| 1513 | 7402 | HLT       |        |                                                 |
| 1514 | 0000 | XXDILX, 0 |        |                                                 |
| 1515 | 6053 | 6053      |        | /LOAD X                                         |
| 1516 | 5714 | JMP I     | XXDILX |                                                 |
| 1517 | 7402 | HLT       |        |                                                 |
| 1520 | 0000 | XXDILY, 0 |        |                                                 |
| 1521 | 6054 | 6054      |        | /LOAD Y                                         |
| 1522 | 5720 | JMP I     | XXDILY |                                                 |
| 1523 | 7402 | HLT       |        |                                                 |
| 1524 | 0000 | XXDIXY, 0 |        |                                                 |
| 1525 | 6055 | 6055      |        | /INTENSIFY                                      |
| 1526 | 5724 | JMP I     | XXDIXY |                                                 |
| 1527 | 7402 | HLT       |        |                                                 |
| 1530 | 0000 | XXDILE, 0 |        |                                                 |
| 1531 | 6056 | 6056      |        | /LOAD ENABLE FROM AC, CLEAR AC                  |
| 1532 | 5732 | JMP I     | XXDILE |                                                 |
| 1533 | 7402 | HLT       |        |                                                 |

1534 0000 /PRINT ROUTINE  
PRLP, 0

|      |      |       |      |                 |
|------|------|-------|------|-----------------|
| 1535 | 6046 | TLS   |      | /XMIT CHARACTER |
| 1536 | 6041 | TSF   |      | /WAIT FOR FLAG  |
| 1537 | 5336 | JMP   | .-1  |                 |
| 1540 | 7200 | CLA   |      |                 |
| 1541 | 5734 | JMP I | PRLP | /RETURN         |

/CARRIAGE RETURN LINE FEED ROUTINE

|      |      |       |      |                       |
|------|------|-------|------|-----------------------|
| 1542 | 0000 | CRLF, | 0    |                       |
| 1543 | 7240 | CLA   | CMA  |                       |
| 1544 | 0042 | AND   | K215 | /CARRIAGE RETURN CODE |
| 1545 | 4334 | JMS   | PRLP | /PRINT ROUTINE        |
| 1546 | 7240 | CLA   | CMA  |                       |
| 1547 | 0041 | AND   | K212 | /LINE FEED CODE       |
| 1550 | 4334 | JMS   | PRLP | /PRINT ROUTINE        |
| 1551 | 5742 | JMP I | CRLF | /RETURN               |

/ROUTINE TO CLEAR WORKING BUFFERS PRIOR TO TEST

|      |      |         |        |                  |
|------|------|---------|--------|------------------|
| 1552 | 0000 | XSETUP, | 0      |                  |
| 1553 | 4451 | JMS I   | XMOVE  | /CLEAR WORK AREA |
| 1554 | 0026 | TEMP0   |        |                  |
| 1555 | 0027 | TEMPA   |        |                  |
| 1556 | 7773 | -5      |        |                  |
| 1557 | 6002 | IOF     |        |                  |
| 1560 | 6007 | CAF     |        |                  |
| 1561 | 1167 | TAD     | [5402  |                  |
| 1562 | 3001 | DCA     | 1      |                  |
| 1563 | 7040 | CMA     |        |                  |
| 1564 | 3061 | DCA     | ERSWIT |                  |
| 1565 | 3767 | DCA I   | XIND   |                  |
| 1566 | 5752 | JMP I   | XSETUP |                  |
| 1567 | 1346 | XIND,   | IND    |                  |

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/ROUTINE TO CHECK IF TEST COMPLETED ITERATION

|      |      |       |        |                      |
|------|------|-------|--------|----------------------|
| 1600 | 0000 | XTAL, | 0      |                      |
| 1601 | 7604 | LAS   |        |                      |
| 1602 | 0022 | AND   | SW2    | /LOOP OVERRIDE?      |
| 1603 | 7640 | SZA   | CLA    |                      |
| 1604 | 5230 | JMP   | XTAL1  | /YES                 |
| 1605 | 7604 | LAS   |        |                      |
| 1606 | 0025 | AND   | SW5    | /TEST SELECTED?      |
| 1607 | 7640 | SZA   | CLA    |                      |
| 1610 | 5214 | JMP   | .,+4   |                      |
| 1611 | 2033 | ISZ   | TALLY  | /DONE WITH TEST?     |
| 1612 | 7410 | SKP   |        | /NO                  |
| 1613 | 5230 | JMP   | XTAL1  | /YES                 |
| 1614 | 1061 | TAD   | ERSWIT | /CHECK FOR ERROR     |
| 1615 | 7640 | SZA   | CLA    | /ERROR THIS PASS?    |
| 1616 | 5224 | JMP   | .,+6   | /NO                  |
| 1617 | 1017 | TAD   | MSGPNT | /GET MESSAGE POINTER |

|      |      |            |        |                          |
|------|------|------------|--------|--------------------------|
| 1620 | 1044 | TAD        | M1     | /DECREMENT POINTER       |
| 1621 | 3017 | DCA        | MSGPNT | /RESTORE POINTER         |
| 1622 | 1044 | TAD        | M1     |                          |
| 1623 | 3061 | DCA        | ERSWIT | /RESTORE ERROR INDICATOR |
| 1624 | 1200 | TAD        | XTAL   | /SET RETURN ADDRESS      |
| 1625 | 1045 | TAD        | M2     |                          |
| 1626 | 3200 | DCA        | XTAL   | /STORE RETURN ADDRESS    |
| 1627 | 5600 | JMP I      | XTAL   |                          |
| 1630 | 2017 | XTAL1, ISZ | MSGPNT |                          |
| 1631 | 5620 | JMP I      | XTAL   |                          |

/POINTER FOR SELECTED TEST OPTION

|      |      |               |
|------|------|---------------|
| 1632 | 0223 | XTEST, TST0-1 |
| 1633 | 0240 | TST1-1        |
| 1634 | 0265 | TST2-1        |
| 1635 | 0306 | TST3-1        |
| 1636 | 0326 | TST4-1        |
| 1637 | 0346 | TST5-1        |
| 1640 | 0400 | TST6-1        |
| 1641 | 0426 | TST7-1        |
| 1642 | 0451 | TST10-1       |
| 1643 | 0501 | TST11-1       |
| 1644 | 0600 | TST12-1       |
| 1645 | 0636 | TST13-1       |
| 1646 | 0710 | TST14-1       |

/ROUTINE TO SELECT SPECIFIC LOGIC TEST SUBROUTINE

|      |      |               |           |                          |
|------|------|---------------|-----------|--------------------------|
| 1647 | 0000 | XSELEC, 0     |           |                          |
| 1650 | 7604 | LAS           | /GET TEST |                          |
| 1651 | 0175 | AND           | C17       |                          |
| 1652 | 3026 | DCA           | TEMP0     | /STORE TEST              |
| 1653 | 1026 | TAD           | TEMP0     |                          |
| 1654 | 1044 | TAD           | M1        |                          |
| 1655 | 1166 | TAD           | C140      |                          |
| 1656 | 3017 | DCA           | 17        | /MESSAGE POINTER SET NOW |
| 1657 | 1026 | TAD           | TEMP0     | /GET TEST                |
| 1660 | 1266 | TAD           | JMPLOC    | /DEVELOP POINTER         |
| 1661 | 0050 | AND           | K77       |                          |
| 1662 | 1267 | TAD           | JMPINS    | /DEVELOP INSTRUCTION     |
| 1663 | 3264 | DCA           | JMPPTR    |                          |
| 1664 | 7402 | JMPPTR, HLT   | /DO IT!   |                          |
| 1665 | 7402 | HLT           | /TRAP     |                          |
| 1666 | 1632 | JMPLOC, XTEST |           |                          |
| 1667 | 5600 | JMPINS, 5600  |           |                          |

/ERROR HANDLERS FOR OPEN LOOP TESTS

|      |      |          |     |
|------|------|----------|-----|
| 1670 | 0000 | ERPT1, 0 |     |
| 1671 | 7604 | LAS      |     |
| 1672 | 0020 | AND      | SW0 |
| 1673 | 7710 | SPA CLA  |     |
| 1674 | 5300 | JMP      | ,+4 |

|      |       |         |         |                    |
|------|-------|---------|---------|--------------------|
| 1675 | 4777' | JMS     | MESSAGE |                    |
| 1676 | 4003  | EMSG20  |         |                    |
| 1677 | 4776' | JMS     | CRLF    |                    |
| 1700 | 4775' | JMS     | MESS    |                    |
| 1701 | 4776' | JMS     | CRLF    |                    |
| 1702 | 7604  | LAS     |         |                    |
| 1703 | 0021  | AND     | SW1     | /HALT ON ERROR?    |
| 1704 | 7650  | SNA CLA |         | /SKIP IF YES       |
| 1705 | 5670  | JMP I   | ERPT1   |                    |
| 1706 | 7402  | HLT     |         |                    |
| 1707 | 5774' | JMP     | RESOL   | /RETURN TO ROUTINE |

|      |       |          |         |                    |
|------|-------|----------|---------|--------------------|
| 1710 | 0000  | ERPT2, 0 |         |                    |
| 1711 | 7604  | LAS      |         |                    |
| 1712 | 0020  | AND      | SW0     |                    |
| 1713 | 7710  | SPA CLA  |         |                    |
| 1714 | 5320  | JMP      | .+4     |                    |
| 1715 | 4777' | JMS      | MESSAGE |                    |
| 1716 | 4034  | EMSG21   |         |                    |
| 1717 | 4776' | JMS      | CRLF    |                    |
| 1720 | 7604  | LAS      |         |                    |
| 1721 | 0021  | AND      | SW1     | /HALT ON ERROR?    |
| 1722 | 7650  | SNA CLA  |         | /SKIP IF YES       |
| 1723 | 5710  | JMP I    | ERPT2   |                    |
| 1724 | 1027  | TAD      | TEMPA   |                    |
| 1725 | 7402  | HLT      |         |                    |
| 1726 | 7200  | CLA      |         |                    |
| 1727 | 1030  | TAD      | TEMPB   |                    |
| 1730 | 7402  | HLT      |         |                    |
| 1731 | 5773' | JMP      | NOISE   | /RETURN TO ROUTINE |

|      |       |          |         |  |
|------|-------|----------|---------|--|
| 1732 | 0000  | ERPT3, 0 |         |  |
| 1733 | 7604  | LAS      |         |  |
| 1734 | 0020  | AND      | SW0     |  |
| 1735 | 7710  | SPA CLA  |         |  |
| 1736 | 5342  | JMP      | .+4     |  |
| 1737 | 4777' | JMS      | MESSAGE |  |
| 1740 | 4056  | EMSG22   |         |  |
| 1741 | 4776' | JMS      | CRLF    |  |
| 1742 | 7604  | LAS      |         |  |
| 1743 | 0021  | AND      | SW1     |  |
| 1744 | 7650  | SNA CLA  |         |  |
| 1745 | 5732  | JMP I    | ERPT3   |  |
| 1746 | 1332  | TAD      | ERPT3   |  |
| 1747 | 1044  | TAD      | M1      |  |
| 1750 | 7402  | HLT      |         |  |

|      |       |          |         |  |
|------|-------|----------|---------|--|
| 1751 | 0000  | ERPT4, 0 |         |  |
| 1752 | 4777' | JMS      | MESSAGE |  |
| 1753 | 4105  | EMSG23   |         |  |
| 1754 | 4776' | JMS      | CRLF    |  |
| 1755 | 5751  | JMP I    | ERPT4   |  |

|      |      |          |  |  |
|------|------|----------|--|--|
| 1756 | 0000 | ERPT5, 0 |  |  |
| 1757 | 7604 | LAS      |  |  |

|      |       |        |         |
|------|-------|--------|---------|
| 1760 | 0020  | AND    | SW0     |
| 1761 | 7710  | SPA    | CLA     |
| 1762 | 5366  | JMP    | ,+4     |
| 1763 | 4777' | JMS    | MESAGE  |
| 1764 | 4122  | EMSG24 |         |
| 1765 | 4776' | JMS    | CRLF    |
| 1766 | 5756  | JMP    | I ERPT5 |

1773 2051  
1774 2200  
1775 3000  
1776 1542  
1777 1274  
2000

PAGE

/MONOTONICITY TEST

|      |      |        |         |                               |
|------|------|--------|---------|-------------------------------|
| 2000 | 7300 | MONOT, | CLA CLL |                               |
| 2001 | 3027 |        | DCA     | TEMPA /CLEAR N AND            |
| 2002 | 3030 |        | DCA     | TEMPB /N+1 CONVERSION STORAGE |
| 2003 | 4520 |        | ADCL    | /CLEAR CONVERTER              |
| 2004 | 4522 |        | ADST    | /START CONVERSION             |
| 2005 | 4524 |        | ADSK    | /WAIT FOR DONE                |
| 2006 | 5205 |        | JMP     | .-1                           |
| 2007 | 4523 |        | ADRB    | /READ A-D BUFFER              |
| 2010 | 3027 |        | DCA     | TEMPA /STORE NTH CONVERSION   |
| 2011 | 7604 | CONT,  | LAS     | /GET SWITCHES                 |
| 2012 | 7040 |        | CMA     | /COMPLEMENT FOR DOWN COUNT    |
| 2013 | 3032 |        | DCA     | CNTR1                         |
| 2014 | 4522 |        | ADST    | /DO N+1ST CONVERSION          |
| 2015 | 4524 |        | ADSK    |                               |
| 2016 | 5215 |        | JMP     | .-1                           |
| 2017 | 4523 |        | ADRB    |                               |
| 2020 | 3030 |        | DCA     | TEMPB /SAVE                   |
| 2021 | 1027 |        | TAD     | TEMPA /SUBTRACT               |
| 2022 | 7041 |        | CIA     |                               |
| 2023 | 1030 |        | TAD     | TEMPB                         |
| 2024 | 7510 |        | SPA     | />0?                          |

|      |       |     |     |                                               |
|------|-------|-----|-----|-----------------------------------------------|
| 2025 | 7041  |     | CIA | /NO, TAKE ABSOLUTE VALUE                      |
| 2026 | 7450  |     | SNA | /DIFFERENCE 0?                                |
| 2027 | 5243  |     | JMP | OK /YES, OK,                                  |
| 2030 | 1044  |     | TAD | M1                                            |
| 2031 | 7650  |     | SNA | CLA /DIFFERENCE = 1?                          |
| 2032 | 5243  |     | JMP | OK /YES, OK,                                  |
| 2033 | 4777' |     | JMS | ERPT4                                         |
| 2034 | 7200  |     | CLA |                                               |
| 2035 | 1027  |     | TAD | TEMPA /DIFFERENCE > 1, DISPLAY NTH CONVERSION |
| 2036 | 7402  |     | HLT |                                               |
| 2037 | 7300  |     | CLA | CLL                                           |
| 2040 | 1030  |     | TAD | TEMPB /DISPLAY N+1 CONVERSION                 |
| 2041 | 7402  |     | HLT |                                               |
| 2042 | 5200  |     | JMP | MONOT /RESTART TO RESYNC                      |
| 2043 | 2032  | OK, | ISZ | CNTR1 /STALL                                  |

|      |      |     |       |                         |
|------|------|-----|-------|-------------------------|
| 2044 | 5243 | JMP | .-1   |                         |
| 2045 | 7300 | CLA | CLL   |                         |
| 2046 | 1030 | TAD | TEMPB | /N+1 CONVERSION BECOMES |
| 2047 | 3027 | DCA | TEMPA | /N                      |
| 2050 | 5211 | JMP | CONT  | /GET N+1 CONVERSION     |

/ROUTINE TO TEST FOR EQUALITY OF TWO SUCCESSIVE ADRB'S.

|      |       |         |         |                         |
|------|-------|---------|---------|-------------------------|
| 2051 | 7300  | NOISE,  | CLA CLL |                         |
| 2052 | 1177  | TAD     | [ -100  | /SET TALLY FOR 64 TIMES |
| 2053 | 3026  | DCA     | TEMP0   |                         |
| 2054 | 1022  | TAD     | SW2     | /ENABLE DONE BIT        |
| 2055 | 4521  | ADLM    |         | /LOAD MPX REG           |
| 2056 | 4522  | ADST    |         | /CONVERT                |
| 2057 | 4524  | ADSK    |         | /DONE FLAG?             |
| 2060 | 5257  | JMP     | .-1     | /NO                     |
| 2061 | 4523  | ADRB    |         | /YES, READ AD BUFFER    |
| 2062 | 3027  | DCA     | TEMPA   | /STORE                  |
| 2063 | 4523  | ADRB    |         | /RE-READ                |
| 2064 | 3030  | DCA     | TEMPB   | /STORE                  |
| 2065 | 1027  | TAD     | TEMPA   | /COMPARE FOR EQUALITY   |
| 2066 | 7041  | CIA     |         |                         |
| 2067 | 1030  | TAD     | TEMPB   |                         |
| 2070 | 7420  | SNL     |         | /LINK SHOULD BE SET     |
| 2071 | 4776' | JMS     | ERPT2   | /NOT EQUAL              |
| 2072 | 7440  | SZA     |         |                         |
| 2073 | 4776' | JMS     | ERPT2   | /NOT EQUAL              |
| 2074 | 7300  | CLA CLL |         |                         |
| 2075 | 2026  | ISZ     | TEMP0   | /CONTINUE               |
| 2076 | 5256  | JMP     | NOISE+5 | /YES                    |
| 2077 | 7200  | CLA     |         |                         |
| 2100 | 1040  | TAD     | K207    |                         |
| 2101 | 4775' | JMS     | PRLP    | /RING BELL              |
| 2102 | 5251  | JMP     | NOISE   | /DO TEST AGAIN          |

/ROUTINE TO CHECK FOR NOISE IN MULTIPLEXER

|      |      |         |         |                             |
|------|------|---------|---------|-----------------------------|
| 2103 | 7300 | GLITCH, | CLA CLL |                             |
| 2104 | 1177 | TAD     | [ -100  |                             |
| 2105 | 3026 | DCA     | TEMP0   |                             |
| 2106 | 7604 | LAS     |         | /OPERATOR TO SELECT CHANNEL |
| 2107 | 0175 | AND     | [17     |                             |
| 2110 | 3031 | DCA     | TEMPC   |                             |
| 2111 | 1031 | TAD     | TEMPC   |                             |
| 2112 | 4521 | ADLM    |         |                             |
| 2113 | 4522 | ADST    |         |                             |
| 2114 | 4524 | ADSK    |         |                             |
| 2115 | 5314 | JMP     | .-1     |                             |
| 2116 | 4523 | ADRB    |         |                             |
| 2117 | 3027 | DCA     | TEMPA   |                             |
| 2120 | 4344 | JMS     | RANCHN  | /GET RANDOM CHANNEL         |
| 2121 | 1077 | TAD     | CHNL    |                             |
| 2122 | 4521 | ADLM    |         |                             |
| 2123 | 4527 | ADRS    |         |                             |
| 2124 | 2026 | ISZ     | TEMP3   |                             |
| 2125 | 5320 | JMP     | CHNL1   |                             |
| 2126 | 7300 | CLA CLL |         |                             |
| 2127 | 4523 | ADRB    |         |                             |
| 2130 | 3030 | DCA     | TEMPB   |                             |
| 2131 | 1027 | TAD     | TEMPA   |                             |

|      |      |     |        |
|------|------|-----|--------|
| 2132 | 7041 | CIA |        |
| 2133 | 1030 | TAD | TEMPB  |
| 2134 | 7420 | SNL |        |
| 2135 | 4774 | JMS | ERPT5  |
| 2136 | 7440 | SZA |        |
| 2137 | 4774 | JMS | ERPT5  |
| 2140 | 7300 | CLA | CLL    |
| 2141 | 1040 | TAD | K207   |
| 2142 | 4775 | JMS | PRLP   |
| 2143 | 5303 | JMP | GLITCH |

|      |      |         |       |        |
|------|------|---------|-------|--------|
| 2144 | 1357 | RANCHN, | TAD   | FSTNO  |
| 2145 | 7006 |         | RTL   |        |
| 2146 | 3357 |         | DCA   | FSTNO  |
| 2147 | 1357 |         | TAD   | FSTNO  |
| 2150 | 1360 |         | TAD   | SECNO  |
| 2151 | 7006 |         | RTL   |        |
| 2152 | 1360 |         | TAD   | SECNO  |
| 2153 | 7012 |         | RTR   |        |
| 2154 | 0175 |         | AND   | E17    |
| 2155 | 3077 |         | DCA   | CHNL   |
| 2156 | 5744 |         | JMP I | RANCHN |

|      |      |        |      |
|------|------|--------|------|
| 2157 | 0437 | FSTNO, | 0437 |
| 2160 | 2525 | SECNO, | 2525 |

2174 1756  
2175 1534  
2176 1710  
2177 1751  
2200

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/ROUTINE TO PERFORM 64 CONVERSIONS OF ANY GIVEN VOLTAGE ON SELECTED CHANNEL

2200 0065 RESOL, SETUP  
2201 1053 TAD XSTOR  
2202 3010 DCA 10  
2203 3270 DCA STORAG  
2204 4451 JMS I XMOVE /CLEAR WORK AREA  
2205 2270 STORAG  
2206 2271 STORAG+1  
2207 7700 -100  
2210 1177 TAD [-100  
2211 3026 DCA TEMP0  
2212 4520 ADCL  
2213 7604 LAS /GET CHANNEL  
2214 0175 AND [17  
2215 3062 DCA CHAN /STORE CHANNEL  
2216 1062 TAD CHAN  
2217 4521 ADLM /LOAD CHANNEL  
2220 4522 ADST  
2221 4524 ADSK  
2222 5221 JMP ,-1  
2223 4523 ADRB  
2224 3410 DCA I 10 /PLACE IN TABLE  
2225 2026 ISZ TEMP0 /DONE?  
2226 5220 JMP ,-6 /NO  
2227 5454 JMP I XCOMPR /YES, NOW CHECK

/STORAGE TABLE FOR VOLTAGE COMPARISONS

2270 \*.50

2270 0000 STORAG, 0 /100(8) LOCATIONS  
2400 PAGE

/ROUTINE TO COMPARE FOR GREATER THAN + OR - 1 LSB DIFFERENCE IN 64 CONVERSIONS

2400 7320 COMPAR, CLA CLL  
2401 1165 TAD [-77  
2402 3026 DCA TEMP0  
2403 1053 TAD XSTOR /POINTER FOR FIRST WORD  
2404 3010 DCA 10  
2405 1410 TAD I 10  
2406 3027 DCA TEMPA  
2407 7220 COMPR1, CLA  
2410 1410 TAD I 10  
2411 3030 DCA TEMPB  
2412 1027 TAD TEMPA  
2413 7041 CIA  
2414 1030 TAD TEMPB  
2415 7440 SZA /SKIP HERE

|      |      |         |        |                                       |
|------|------|---------|--------|---------------------------------------|
| 2416 | 5222 | JMP     | ,+4    | /AND                                  |
| 2417 | 7420 | SNL     |        | /HERE IF =                            |
| 2420 | 5222 | JMP     | ,+2    |                                       |
| 2421 | 5257 | JMP     | AOK    |                                       |
| 2422 | 7430 | SZL     |        |                                       |
| 2423 | 5230 | JMP     | ,+5    |                                       |
| 2424 | 7040 | CMA     |        |                                       |
| 2425 | 7440 | SZA     |        | /SKIP HERE IF DIFFERENCE +1 LSB       |
| 2426 | 7410 | SKP     |        |                                       |
| 2427 | 5257 | JMP     | AOK    |                                       |
| 2430 | 7100 | CLL     |        |                                       |
| 2431 | 7010 | RAR     |        |                                       |
| 2432 | 7440 | SZA     |        | /SKIP HERE                            |
| 2433 | 5237 | JMP     | ,+4    | /AND                                  |
| 2434 | 7420 | SNL     |        | /HERE IF DIFFERENCE -1 LSB            |
| 2435 | 7410 | SKP     |        |                                       |
| 2436 | 5257 | JMP     | AOK    |                                       |
| 2437 | 7300 | CLA CLL |        | /CHECK FOR SPECIAL CASE OF 7777 AND 0 |
| 2440 | 1027 | TAD     | TEMPA  |                                       |
| 2441 | 7440 | SZA     |        | /A=0?                                 |
| 2442 | 7410 | SKP     |        | /NO                                   |
| 2443 | 5247 | JMP     | ,+4    | /YES                                  |
| 2444 | 7040 | CMA     |        | /A=7777?                              |
| 2445 | 7440 | SZA     |        | /SKIP IF YES                          |
| 2446 | 4777 | JMS     | ERPT1  |                                       |
| 2447 | 1030 | TAD     | TEMPB  | /A = 7777 OR 0                        |
| 2450 | 7440 | SZA     |        | /B=0?                                 |
| 2451 | 5253 | JMP     | ,+2    | /NO                                   |
| 2452 | 5257 | JMP     | AOK    |                                       |
| 2453 | 7040 | CMA     |        | /B=7777?                              |
| 2454 | 7440 | SZA     |        | /SKIP IF YES                          |
| 2455 | 4777 | JMS     | ERPT1  |                                       |
| 2456 | 5257 | JMP     | AOK    |                                       |
| 2457 | 7300 | CLA CLL |        |                                       |
| 2460 | 1030 | TAD     | TEMPB  |                                       |
| 2461 | 3027 | DCA     | TEMPA  |                                       |
| 2462 | 2026 | ISZ     | TEMP0  | /DONE?                                |
| 2463 | 5207 | JMP     | COMPR1 | /NO                                   |
| 2464 | 5776 | JMP     | RESOL  | /YES, REPEAT TEST                     |

2576 2200  
2577 1670  
2600

PAGE

/LABB-E SYSTEM CHECK

|      |      |       |       |                        |
|------|------|-------|-------|------------------------|
| 2600 | 0000 | SYST, | 0     |                        |
| 2601 | 4465 | JMS I | SETUP |                        |
| 2602 | 4520 | ADCL  |       |                        |
| 2603 | 7402 | HLT   |       |                        |
| 2604 | 7604 | LAS   |       |                        |
| 2605 | 0377 | AND   | (730  |                        |
| 2606 | 1376 | TAD   | (4040 | /RATE AND ENABLE EXT'L |
| 2607 | 3031 | DCA   | TEMPC | /SAVE                  |
| 2610 | 1031 | TAD   | TEMPC |                        |

|      |      |         |         |                                 |
|------|------|---------|---------|---------------------------------|
| 2611 | 4530 | CLOE    |         |                                 |
| 2612 | 7040 | CMA     |         |                                 |
| 2613 | 4532 | CLZE    |         |                                 |
| 2614 | 7200 | CLA     |         |                                 |
| 2615 | 1024 | TAD     | SW4     | /EXT START FOR A-D              |
| 2616 | 3026 | DCA     | TEMP2   |                                 |
| 2617 | 4775 | JMS     | MESSAGE |                                 |
| 2620 | 4215 | AUTMSG  |         |                                 |
| 2621 | 7402 | HLT     |         |                                 |
| 2622 | 7604 | LAS     |         |                                 |
| 2623 | 0025 | AND     | SW5     |                                 |
| 2624 | 7440 | SZA     |         | /SKIP IF NOT AUTO-INCREMENT     |
| 2625 | 4321 | JMS     | LSTCHN  | /CHECK FOR LAST CHANNEL         |
| 2626 | 7604 | LAS     |         |                                 |
| 2627 | 0175 | AND     | [17     |                                 |
| 2630 | 4521 | ADLM    |         | /LOAD CHANNEL                   |
| 2631 | 1026 | TAD     | TEMP0   |                                 |
| 2632 | 4526 | ADLE    |         | /LOAD EXT ENABLE BIT IF PRESENT |
| 2633 | 1026 | TAD     | TEMP0   |                                 |
| 2634 | 7650 | SNA CLA |         | /SKIP FOR EXTL ENABLE           |
| 2635 | 5245 | JMP     | ,+10    |                                 |
| 2636 | 1374 | TAD     | (7001   | /-X(MAX)                        |
| 2637 | 3027 | DCA     | TEMPA   |                                 |
| 2640 | 4533 | CLSA    |         |                                 |
| 2641 | 4531 | CLSK    |         |                                 |
| 2642 | 5241 | JMP     | , -1    |                                 |
| 2643 | 7240 | CLA CMA |         |                                 |
| 2644 | 4532 | CLZE    |         | /STOP CLOCK                     |
| 2645 | 7200 | CLA     |         |                                 |
| 2646 | 7410 | SKP     |         |                                 |
| 2647 | 4522 | ADST    |         |                                 |
| 2650 | 4524 | ADSK    |         |                                 |
| 2651 | 5250 | JMP     | , -1    |                                 |
| 2652 | 4527 | ADRS    |         |                                 |
| 2653 | 0175 | AND     | [17     |                                 |
| 2654 | 1030 | TAD     | TEMPB   |                                 |
| 2655 | 7001 | IAC     |         |                                 |
| 2656 | 7440 | SZA     |         |                                 |
| 2657 | 5261 | JMP     | , +2    |                                 |
| 2660 | 4521 | ADLM    |         |                                 |
| 2661 | 4523 | ADRB    |         |                                 |
| 2662 | 4540 | DILY    |         |                                 |
| 2663 | 7200 | CLA     |         |                                 |
| 2664 | 1027 | TAD     | TEMPA   |                                 |
| 2665 | 4537 | DILX    |         |                                 |
| 2666 | 7001 | IAC     |         |                                 |
| 2667 | 3027 | DCA     | TEMPA   |                                 |
| 2670 | 1027 | TAD     | TEMPA   |                                 |
| 2671 | 1374 | TAD     | (7001   |                                 |
| 2672 | 7640 | SZA CLA |         | /SKIP IF +X(MAX)                |
| 2673 | 7410 | SKP     |         |                                 |
| 2674 | 5325 | JMP     | RESTR   |                                 |
| 2675 | 4536 | DISD    |         |                                 |
| 2676 | 5275 | JMP     | , -1    |                                 |

|      |      |            |         |                    |
|------|------|------------|---------|--------------------|
| 2677 | 4541 | DIXY       |         |                    |
| 2700 | 1046 | TAD        | M4      |                    |
| 2701 | 3340 | DCA        | TEMPX   |                    |
| 2702 | 2340 | ISZ        | TEMPX   |                    |
| 2703 | 5302 | JMP        | , -1    |                    |
| 2704 | 5245 | JMP        | CLKST+7 |                    |
| 2705 | 1031 | RESTR, TAD | TEMPC   | /TO RESTART CLOCK  |
| 2706 | 4530 | CLOE       |         |                    |
| 2707 | 7040 | CMA        |         |                    |
| 2710 | 4532 | CLZE       |         |                    |
| 2711 | 7604 | LAS        |         |                    |
| 2712 | 0025 | AND        | SW5     | /A-I MODE          |
| 2713 | 7640 | SZA CLA    |         | /SKIP IF NO        |
| 2714 | 5236 | JMP        | CLKST   |                    |
| 2715 | 7604 | LAS        |         |                    |
| 2716 | 0175 | AND        | [17     | /TO CHANGE CHANNEL |
| 2717 | 4521 | ADLM       |         |                    |
| 2720 | 5236 | JMP        | CLKST   | /GO                |
| 2721 | 0000 | LSTCHN, 0  |         |                    |
| 2722 | 7604 | LAS        |         |                    |
| 2723 | 0175 | AND        | [17     |                    |
| 2724 | 7040 | CMA        |         |                    |
| 2725 | 3030 | DCA        | TEMPB   |                    |
| 2726 | 2321 | ISZ        | LSTCHN  |                    |
| 2727 | 2321 | ISZ        | LSTCHN  |                    |
| 2730 | 7604 | LAS        |         |                    |
| 2731 | 0025 | AND        | SW5     |                    |
| 2732 | 7650 | SNA CLA    |         |                    |
| 2733 | 5337 | JMP        | , +4    |                    |
| 2734 | 1024 | TAD        | SW4     |                    |
| 2735 | 1025 | TAD        | SW5     |                    |
| 2736 | 3026 | DCA        | TEMP0   |                    |
| 2737 | 5721 | JMP I      | LSTCHN  |                    |
| 2740 | 0000 | TEMPX, 0   |         |                    |

|      |      |         |       |  |
|------|------|---------|-------|--|
| 2774 | 7021 |         |       |  |
| 2775 | 1274 |         |       |  |
| 2776 | 4040 |         |       |  |
| 2777 | 0700 |         |       |  |
|      | 3000 |         |       |  |
| 3000 | 0000 | PAGE    |       |  |
|      |      | MESS, 0 |       |  |
| 3001 | 4777 | JMS     | CRLF  |  |
| 3002 | 7300 | CLA CLL |       |  |
| 3003 | 1027 | TAD     | TEMPA |  |
| 3004 | 2376 | AND     | (7000 |  |
| 3005 | 7022 | BSW     |       |  |
| 3006 | 7012 | RTR     |       |  |
| 3007 | 7010 | RAR     |       |  |
| 3010 | 1375 | TAD     | (260  |  |
| 3011 | 4774 | JMS     | PRLP  |  |
| 3012 | 7300 | CLA CLL |       |  |
| 3013 | 1027 | TAD     | TEMPA |  |
| 3014 | 7006 | RTL     |       |  |
| 3015 | 7024 | RAL     |       |  |

|      |       |         |       |
|------|-------|---------|-------|
| 3016 | 2376  | AND     | (7000 |
| 3017 | 7022  | BSW     |       |
| 3020 | 7012  | RTR     |       |
| 3021 | 7010  | RAR     |       |
| 3022 | 1375  | TAD     | (260  |
| 3023 | 4774' | JMS     | PRLP  |
| 3024 | 7320  | CLA CLL |       |
| 3025 | 1027  | TAD     | TEMPA |
| 3026 | 7012  | RTR     |       |
| 3027 | 7010  | RAR     |       |
| 3030 | 0373  | AND     | (7    |
| 3031 | 1375  | TAD     | (260  |
| 3032 | 4774' | JMS     | PRLP  |
| 3033 | 7320  | CLA CLL |       |
| 3034 | 1027  | TAD     | TEMPA |
| 3035 | 0373  | AND     | (7    |
| 3036 | 1375  | TAD     | (260  |
| 3037 | 4774' | JMS     | PRLP  |
| 3040 | 7320  | CLA CLL |       |
| 3041 | 4777' | JMS     | CRLF  |
| 3042 | 7320  | CLA CLL |       |
| 3043 | 1030  | TAD     | TEMPB |
| 3044 | 0376  | AND     | (7000 |
| 3045 | 7022  | BSW     |       |
| 3046 | 7010  | RAR     |       |
| 3047 | 7012  | RTR     |       |
| 3050 | 1375  | TAD     | (260  |
| 3051 | 4774' | JMS     | PRLP  |
| 3052 | 7320  | CLA CLL |       |
| 3053 | 1030  | TAD     | TEMPB |
| 3054 | 7006  | RTL     |       |
| 3055 | 7004  | RAL     |       |
| 3056 | 0376  | AND     | (7000 |
| 3057 | 7022  | BSW     |       |
| 3060 | 7010  | RAR     |       |
| 3061 | 7012  | RTR     |       |
| 3062 | 1375  | TAD     | (260  |
| 3063 | 4774' | JMS     | PRLP  |
| 3064 | 7320  | CLA CLL |       |
| 3065 | 1030  | TAD     | TEMPB |
| 3066 | 7010  | RAR     |       |
| 3067 | 7012  | RTR     |       |
| 3070 | 0373  | AND     | (7    |
| 3071 | 1375  | TAD     | (260  |
| 3072 | 4774' | JMS     | PRLP  |
| 3073 | 7320  | CLA CLL |       |
| 3074 | 1030  | TAD     | TEMPB |
| 3075 | 0373  | AND     | (7    |
| 3076 | 1375  | TAD     | (260  |
| 3077 | 4774' | JMS     | PRLP  |
| 3100 | 7320  | CLA CLL |       |
| 3101 | 4777' | JMS     | CRLF  |
| 3102 | 4777' | JMS     | CRLF  |
| 3103 | 7320  | CLA CLL |       |
| 3104 | 5600  | JMP I   | MESS  |

3173 0007  
 3174 1534  
 3175 0260  
 3176 7000  
 3177 1542  
 3200

PAGE

/CONTROL LOGIC ERROR MESSAGES

3200 3736  
 3201 2405  
 3202 2524  
 3203 4060  
 3204 4055  
 3205 4004  
 3206 1716  
 3207 0540  
 3210 0614  
 3211 0107  
 3212 4017  
 3213 2240  
 3214 2411  
 3215 1511  
 3216 1607  
 3217 4005  
 3220 2222  
 3221 1722  
 3222 4006  
 3223 1401  
 3224 0740  
 3225 1617  
 3226 2440  
 3227 0314  
 3230 0501  
 3231 2205  
 3232 0440  
 3233 1722  
 3234 4023  
 3235 1311  
 3236 2040  
 3237 0601  
 3240 1114  
 3241 2522  
 3242 0537  
 3243 3600  
 3244 3736  
 3245 2405  
 3246 2524  
 3247 4061  
 3250 4055  
 3251 4004  
 3252 1716  
 3253 0540

EMSG2, TEXT "♦♦TEST 0 - DONE FLAG OR TIMING ERROR FLAG NOT CLEARED OR SKIP FAILURE♦♦"

EMSG1, TEXT "♦♦TEST 1 - DONE FLAG NOT SET THEN CLEARED OR SKIP FAILURE♦♦"

3254 0614  
3255 0107  
3256 4016  
3257 1724  
3260 4023  
3261 0524  
3262 4024  
3263 1005  
3264 1640  
3265 0314  
3266 0501  
3267 2205  
3270 0440  
3271 1722  
3272 4023  
3273 1311  
3274 2040  
3275 0601  
3276 1114  
3277 2522  
3300 0537  
3301 3600  
3302 3736  
3303 2405  
3304 2324  
3305 4062  
3306 4055  
3307 4024  
3310 1115  
3311 1116  
3312 0740  
3313 0522  
3314 2217  
3315 2240  
3316 0614  
3317 0107  
3320 4016  
3321 1724  
3322 4023  
3323 0524  
3324 4024  
3325 1005  
3326 1640  
3327 0314  
3330 0501  
3331 2205  
3332 0440  
3333 1722  
3334 4023  
3335 1311  
3336 2040  
3337 0601  
3340 1114  
3341 2522  
3342 0537

EMSG2, TEXT "••TEST 2 - TIMING ERROR FLAG NOT SET THEN CLEARED OR SKIP FAILURE••"

3343 3600  
 3344 3736  
 3345 2405  
 3346 2324  
 3347 4063  
 3350 4055  
 3351 4025  
 3352 1605  
 3353 3020  
 3354 0503  
 3355 2405  
 3356 0440  
 3357 1116  
 3360 2405  
 3361 2222  
 3362 2520  
 3363 2440  
 3364 1703  
 3365 0325  
 3366 2222  
 3367 0504  
 3370 3736  
 3371 0000  
 3372 3736  
 3373 2405  
 3374 2324  
 3375 4064  
 3376 4055  
 3377 4001  
 3400 0422  
 3401 0240  
 3402 0601  
 3403 1114  
 3404 0504  
 3405 4024  
 3406 1740  
 3407 1201  
 3410 1540  
 3411 2422  
 3412 0116  
 3413 2306  
 3414 0522  
 3415 4024  
 3416 1740  
 3417 0103  
 3420 3736  
 3421 0000  
 3422 3736  
 3423 2405  
 3424 2324  
 3425 4065  
 3426 4055  
 3427 4001  
 3430 0422  
 3431 2340

EMSG3, TEXT "++TEST 3 - UNEXPECTED INTERRUPT OCCURRED++"

EMSG4, TEXT "++TEST 4 - ADRB FAILED TO JAM TRANSFER TO AC++"

EMSG5, TEXT "++TEST 5 - ADRS FAILED TO JAM TRANSFER TO AC++"

3432 0621  
3433 1114  
3434 0524  
3435 4024  
3436 1740  
3437 1201  
3440 1540  
3441 2422  
3442 0116  
3443 2326  
3444 0522  
3445 4024  
3446 1740  
3447 0103  
3450 3736  
3451 0000  
3452 3736  
3453 2405  
3454 2324  
3455 4066  
3456 4055  
3457 4005  
3460 1601  
3461 0214  
3462 0540  
3463 2205  
3464 0711  
3465 2324  
3466 0522  
3467 4016  
3470 1724  
3471 4020  
3472 2217  
3473 2005  
3474 2214  
3475 3140  
3476 1417  
3477 0104  
3500 0504  
3501 3736  
3502 0000  
3503 3736  
3504 2405  
3505 2324  
3506 4067  
3507 4055  
3510 4006  
3511 0111  
3512 1405  
3513 0440  
3514 2417  
3515 4007  
3516 0516  
3517 0522  
3520 0124

EMSG6, TEXT "++TEST 6 - ENABLE REGISTER NOT PROPERLY LOADED++"

EMSG7, TEXT "++TEST 7 - FAILED TO GENERATE INTERRUPT WITH DONE FLAG++"

3521 0540  
 3522 1116  
 3523 2405  
 3524 2222  
 3525 2520  
 3526 2440  
 3527 2711  
 3530 2410  
 3531 4024  
 3532 1716  
 3533 0540  
 3534 0614  
 3535 0107  
 3536 3736  
 3537 0000  
 3540 3736  
 3541 2405  
 3542 2324  
 3543 4061  
 3544 6040  
 3545 5540  
 3546 0601  
 3547 1114  
 3550 0504  
 3551 4024  
 3552 1740  
 3553 0705  
 3554 1605  
 3555 2201  
 3556 2405  
 3557 4011  
 3560 1624  
 3561 0522  
 3562 2225  
 3563 2024  
 3564 4027  
 3565 1124  
 3566 1040  
 3567 2411  
 3570 1511  
 3571 1607  
 3572 4005  
 3573 2222  
 3574 1722  
 3575 4006  
 3576 1401  
 3577 0737  
 3600 3620  
 3601 3736  
 3602 2405  
 3603 2324  
 3604 4061  
 3605 6140  
 3606 5540  
 3607 0601

MSG10, TEXT "••TEST 10 - FAILED TO GENERATE INTERRUPT WITH TIMING ERROR FLAG••"

MSG11, TEXT "••TEST 11 - FAILED TO LOAD AND READ MPX REG AND CLEAR AC••"

3610 1114  
3611 0504  
3612 4024  
3613 1740  
3614 1417  
3615 0104  
3616 4001  
3617 1604  
3620 4022  
3621 0501  
3622 0440  
3623 1520  
3624 3040  
3625 2205  
3626 0740  
3627 0116  
3630 0440  
3631 0314  
3632 0501  
3633 2240  
3634 0103  
3635 3736  
3636 0000  
3637 3736  
3640 2405  
3641 2324  
3642 4061  
3643 6240  
3644 5040  
3645 0601  
3646 1114  
3647 0504  
3650 4024  
3651 1740  
3652 1417  
3653 0104  
3654 4001  
3655 1604  
3656 4022  
3657 0501  
3660 0440  
3661 0114  
3662 1440  
3663 0310  
3664 0116  
3665 1605  
3666 1423  
3667 4011  
3670 1624  
3671 1740  
3672 1520  
3673 3040  
3674 2205  
3675 0737  
3676 3600

EMSG12, TEXT "••TEST 12 - FAILED TO LOAD AND READ ALL CHANNELS INTO MPX REG••"

3677 3736  
 3700 2425  
 3701 2324  
 3702 4061  
 3703 6340  
 3704 5540  
 3705 0601  
 3706 1114  
 3707 0504  
 3710 4024  
 3711 1740  
 3712 1417  
 3713 0104  
 3714 4001  
 3715 1604  
 3716 4022  
 3717 0501  
 3720 0440  
 3721 0114  
 3722 1440  
 3723 0310  
 3724 0116  
 3725 1605  
 3726 1423  
 3727 4011  
 3730 1640  
 3731 0125  
 3732 2417  
 3733 5511  
 3734 1603  
 3735 2205  
 3736 1505  
 3737 1624  
 3740 4015  
 3741 1704  
 3742 0537  
 3743 3600  
 3744 3736  
 3745 2405  
 3746 2324  
 3747 4061  
 3750 6440  
 3751 5540  
 3752 0601  
 3753 1114  
 3754 0504  
 3755 4024  
 3756 1740  
 3757 0317  
 3760 1520  
 3761 1405  
 3762 2425  
 3763 4003  
 3764 1716  
 3765 2605

MSG13, TEXT "++TEST 13 - FAILED TO LOAD AND READ ALL CHANNELS IN AUTO-INCREMENT MODE++"

MSG14, TEXT "++TEST 14 - FAILED TO COMPLETE CONVERSION IN SPECIFIED TIME++"

3766 2223  
 3767 1117  
 3770 1640  
 3771 1116  
 3772 4023  
 3773 2005  
 3774 0311  
 3775 0611  
 3776 0504  
 3777 4024  
 4000 1115  
 4001 0537  
 4002 3600  
 4003 3736  
 4004 0601  
 4005 1114  
 4006 0504  
 4007 4024  
 4010 1740  
 4011 2205  
 4012 2317  
 4013 1426  
 4014 0540  
 4015 0317  
 4016 1626  
 4017 0522  
 4020 2311  
 4021 1716  
 4022 2340  
 4023 2417  
 4024 4053  
 4025 4017  
 4026 2240  
 4027 5540  
 4030 6140  
 4031 1423  
 4032 0237  
 4033 3600  
 4034 3736  
 4035 2427  
 4036 1740  
 4037 2525  
 4040 0303  
 4041 0523  
 4042 2311  
 4043 2605  
 4044 4022  
 4045 0501  
 4046 0423  
 4047 4016  
 4050 1724  
 4051 4025  
 4052 2125  
 4053 0114  
 4054 3736

EMSG20, TEXT "••FAILED TO RESOLVE CONVERSIONS TO + OR - 1 LSB••"

EMSG21, TEXT "••TWO SUCCESSIVE READS NOT EQUAL••"

|      |      |                                                                 |
|------|------|-----------------------------------------------------------------|
| 4055 | 0000 |                                                                 |
| 4056 | 3736 | EMSG22, TEXT    "••ERRONEOUS EXTERNAL ENABLE OR TIMING ERROR••" |
| 4057 | 0522 |                                                                 |
| 4060 | 2217 |                                                                 |
| 4061 | 1605 |                                                                 |
| 4062 | 1725 |                                                                 |
| 4063 | 2340 |                                                                 |
| 4064 | 0530 |                                                                 |
| 4065 | 2405 |                                                                 |
| 4066 | 2216 |                                                                 |
| 4067 | 0114 |                                                                 |
| 4070 | 4005 |                                                                 |
| 4071 | 1601 |                                                                 |
| 4072 | 0214 |                                                                 |
| 4073 | 0540 |                                                                 |
| 4074 | 1722 |                                                                 |
| 4075 | 4024 |                                                                 |
| 4076 | 1115 |                                                                 |
| 4077 | 1116 |                                                                 |
| 4100 | 0740 |                                                                 |
| 4101 | 0522 |                                                                 |
| 4102 | 2217 |                                                                 |
| 4103 | 2237 |                                                                 |
| 4104 | 3600 |                                                                 |
| 4105 | 3736 | EMSG23, TEXT    "••MONOTONICITY FAILURE••"                      |
| 4106 | 1517 |                                                                 |
| 4107 | 1617 |                                                                 |
| 4110 | 2411 |                                                                 |
| 4111 | 1611 |                                                                 |
| 4112 | 0311 |                                                                 |
| 4113 | 2431 |                                                                 |
| 4114 | 4006 |                                                                 |
| 4115 | 0111 |                                                                 |
| 4116 | 1425 |                                                                 |
| 4117 | 2205 |                                                                 |
| 4120 | 3736 |                                                                 |
| 4121 | 0000 |                                                                 |
| 4122 | 3736 | EMSG24, TEXT    "••NOISE IN MULTIPLEXER AND A-D BUFFER••"       |
| 4123 | 1617 |                                                                 |
| 4124 | 1123 |                                                                 |
| 4125 | 0540 |                                                                 |
| 4126 | 1116 |                                                                 |
| 4127 | 4015 |                                                                 |
| 4130 | 2514 |                                                                 |
| 4131 | 2411 |                                                                 |
| 4132 | 2014 |                                                                 |
| 4133 | 0530 |                                                                 |
| 4134 | 0522 |                                                                 |
| 4135 | 4001 |                                                                 |
| 4136 | 1604 |                                                                 |
| 4137 | 4001 |                                                                 |
| 4140 | 5504 |                                                                 |
| 4141 | 4002 |                                                                 |
| 4142 | 2506 |                                                                 |
| 4143 | 2605 |                                                                 |

4144 2237  
 4145 3620  
 4146 3736  
 4147 0516  
 4150 0440  
 4151 1706  
 4152 4014  
 4153 1707  
 4154 1103  
 4155 4024  
 4156 0523  
 4157 2437  
 4160 3620

/END OF LOGIC TEST TYPESTRING  
 XEND, TEXT "END OF LOGIC TEST"

4161 3736  
 4162 0104  
 4163 7005  
 4164 4001  
 4165 4024  
 4166 1740  
 4167 0440  
 4170 0317  
 4171 1626  
 4172 0522  
 4173 2405  
 4174 2254  
 4175 4001  
 4176 1570  
 4177 0540  
 4200 1525  
 4201 1424  
 4202 1120  
 4203 1405  
 4204 3005  
 4205 2240  
 4206 0411  
 4207 0107  
 4210 1617  
 4211 2324  
 4212 1103  
 4213 3736  
 4214 0000  
 4215 3736  
 4216 2305  
 4217 2440  
 4220 2527  
 4221 6540  
 4222 5001  
 4223 2524  
 4224 1755  
 4225 1116  
 4226 0351  
 4227 5440  
 4230 4340

/HEADER MESSAGE

XLABEL, TEXT "AD8E A TO D CONVERTER, AM8E MULTIPLEXER DIAGNOSTIC"

AUTMSG, TEXT "SET SW5 (AUTO-INC), # OF CHANS IN SW8-11, OR SET SW8-11 (SINGLE CHAN)"

4231 1706  
 4232 4003  
 4233 1001  
 4234 1623  
 4235 4011  
 4236 1640  
 4237 2327  
 4240 7055  
 4241 6161  
 4242 5440  
 4243 1722  
 4244 4023  
 4245 0524  
 4246 4023  
 4247 2770  
 4250 5561  
 4251 6140  
 4252 5023  
 4253 1116  
 4254 0714  
 4255 0540  
 4256 0310  
 4257 0116  
 4260 5137  
 4261 3600

\$

0165 7701  
 0166 0140  
 0167 5402  
 0170 0076  
 0171 7760  
 0172 0475  
 0173 0445  
 0174 7761  
 0175 0017  
 0176 0322  
 0177 7700

[illegible]

|      |          |          |          |          |          |          |          |          |
|------|----------|----------|----------|----------|----------|----------|----------|----------|
| 4000 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 |
| 4100 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 |

|      |          |          |          |          |          |          |          |          |
|------|----------|----------|----------|----------|----------|----------|----------|----------|
| 4200 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11111111 | 11000000 | 00000000 |
| 4300 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 |

4400  
4500

4600  
4700

5000  
5100

5200  
5300

5400  
5500

5600  
5700

6000  
6100

6200  
6300

6400  
6500

6600  
6700

7000  
7100

7200  
7300

7400  
7500

7600  
7700

|        |      |         |      |        |      |        |      |
|--------|------|---------|------|--------|------|--------|------|
| ADCL   | 4520 | EMSG6   | 3452 | SCOPE  | 1257 | XCONVT | 0035 |
| ADLE   | 4526 | EMSG7   | 3503 | SECNO  | 2160 | XDILE  | 0142 |
| ADLM   | 4521 | EOUT    | 1242 | SELECT | 0064 | XDILX  | 0137 |
| ADRB   | 4523 | ERMSG   | 0145 | SETUP  | 0065 | XDILY  | 0140 |
| ADRS   | 4527 | ERPT1   | 1670 | START  | 0211 | XDISD  | 0136 |
| ADSE   | 4525 | ERPT2   | 1710 | STORAG | 2270 | XDIXY  | 0141 |
| ADSK   | 4524 | ERPT3   | 1732 | SW0    | 0020 | XEND   | 4146 |
| ADST   | 4522 | ERPT4   | 1751 | SW1    | 0021 | XGLIT  | 0057 |
| AOK    | 2457 | ERPT5   | 1756 | SW2    | 0022 | XIND   | 1567 |
| AUTMSG | 4215 | ERR     | 0034 | SW3    | 0023 | XINSTR | 0036 |
| AUTO1  | 0643 | ERSWIT  | 0061 | SW4    | 0024 | XLABEL | 4161 |
| AUTO2  | 0674 | ERTYP   | 1226 | SW5    | 0025 | XMONOT | 0037 |
| BSW    | 7002 | EXIT    | 1271 | SYST   | 2600 | XMOVE  | 0051 |
| CAF    | 6007 | EXT1    | 1032 | TADDR  | 1224 | XNOISE | 0056 |
| CHAN   | 0062 | EXTBL   | 0052 | TAL    | 0063 | XRESOL | 0055 |
| CHNL   | 0077 | EXTL    | 1024 | TALLY  | 0033 | XSELEC | 1647 |
| CHNL1  | 2120 | EXTTE   | 1066 | TEMP0  | 0026 | XSETUP | 1552 |
| CLAB   | 4535 | FADDR   | 1223 | TEMPA  | 0027 | XSTOR  | 0053 |
| CLED   | 4534 | FINIS   | 0730 | TEMPB  | 0030 | XSYST  | 0060 |
| CLKST  | 2636 | FSTNO   | 2157 | TEMPC  | 0031 | XTAL   | 1600 |
| CLOE   | 4530 | GLITCH  | 2103 | TEMPX  | 2740 | XTAL1  | 1630 |
| CLSA   | 4533 | IND     | 1346 | TMG1   | 0475 | XTEST  | 1632 |
| CLSK   | 4531 | INITL   | 0216 | TST0   | 0224 | XXADCL | 1410 |
| CLZE   | 4532 | INSTR   | 1400 | TST1   | 0241 | XXADLE | 1444 |
| CNTR1  | 0032 | JMPINS  | 1667 | TST10  | 0452 | XXADLM | 1414 |
| COMPAR | 2400 | JMPLOC  | 1666 | TST11  | 0502 | XXADRB | 1424 |
| COMPR1 | 2407 | JMPPTR  | 1664 | TST12  | 0601 | XXADRS | 1450 |
| CONT   | 2011 | K1000   | 0047 | TST13  | 0637 | XXADSE | 1436 |
| CONVT  | 1000 | K207    | 0040 | TST14  | 0711 | XXADSK | 1430 |
| CRLF   | 1542 | K212    | 0041 | TST2   | 0266 | XXADST | 1420 |
| DILE   | 4542 | K215    | 0042 | TST3   | 0307 | XXCLAB | 1502 |
| DILX   | 4537 | K6500   | 0043 | TST3S  | 0322 | XXCLED | 1476 |
| DILY   | 4540 | K77     | 0050 | TST4   | 0327 | XXCLOE | 1454 |
| DISD   | 4536 | LSTCHN  | 2721 | TST5   | 0347 | XXCLSA | 1472 |
| DIXY   | 4541 | M1      | 0044 | TST6   | 0401 | XXCLSK | 1460 |
| DON1   | 0445 | M2      | 0045 | TST7   | 0427 | XXCLZE | 1466 |
| EMSG0  | 3200 | M4      | 0046 | TYPECH | 1314 | XXDILE | 1530 |
| EMSG1  | 3244 | MCTR    | 1225 | XADCL  | 0120 | XXDILX | 1514 |
| EMSG10 | 3540 | MESSAGE | 1274 | XADLE  | 0126 | XXDILY | 1520 |
| EMSG11 | 3601 | MESS    | 3000 | XADLM  | 0121 | XXDISD | 1506 |
| EMSG12 | 3637 | MONOT   | 2000 | XADRB  | 0123 | XXDIXY | 1524 |
| EMSG13 | 3677 | MOVE    | 1200 | XADRS  | 0127 |        |      |
| EMSG14 | 3744 | MOVEA   | 1213 | XADSE  | 0125 |        |      |
| EMSG2  | 3302 | MSGPNT  | 0017 | XADSK  | 0124 |        |      |
| EMSG20 | 4003 | MSRGHT  | 1313 | XADST  | 0122 |        |      |
| EMSG21 | 4034 | MTP     | 1340 | XCLAB  | 0135 |        |      |
| EMSG22 | 4056 | NOISE   | 2051 | XCLED  | 0134 |        |      |
| EMSG23 | 4125 | OK      | 2043 | XCLOE  | 0130 |        |      |
| EMSG24 | 4122 | PRLP    | 1534 | XCLSA  | 0133 |        |      |
| EMSG3  | 3344 | RANCHN  | 2144 | XCLSK  | 0131 |        |      |
| EMSG4  | 3372 | RESOL   | 2200 | XCLZE  | 0132 |        |      |
| EMSG5  | 3422 | RESTR   | 2705 | XCOMPR | 0054 |        |      |

ERRORS DETECTED: 0

LINKS GENERATED: 49

RUN-TIME: 14 SECONDS

3K CORE USED

